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Chapter 13: From Scaling Laws to Mapping Chains—The Epistemological Limits and Engineering Principles of AI

Abstract

This chapter applies the fiber framework to the AI domain, organizing the diagnosis in a "4+1+3" structure: mapping chain structure (L0), optimization dynamics (L1), pseudo-calibration (L2), and multi-mapping recursion (L3) respectively examine information transformation, optimization objectives, the appearance of certainty, and dependencies among readings; narrative load multiplication (L4) examines how interaction design and discourse affect the conditions for calibration. Each layer provides a different analytical entry point; this does not mean that every system already exhibits all failure forms. Three transversal dimensions—political economy, physical substrate, and temporal dynamics—cross-cut these layers, with their full unfolding reserved for subsequent chapters.

The chapter unfolds its analysis in the order of mapping structure, optimization objective, output presentation, and dependency among readings, without presenting this as a failure chain that all systems must pass through. When preference-driven optimization lacks independent calibration, the appearance of certainty may mask uncertainties that have yet to be ruled out; when these outputs become the basis for subsequent judgment, evaluation, or training, the risk in turn depends on whether their conditions and uncertainties are preserved and whether independent verification can intervene. Mutual reinforcement between layers can exist—for example, certainty outputs receiving higher preference evaluations, or model-generated content returning into training data—but

the existence of these paths and the strength of their consequences must be judged separately.

The object diagnosed by the fiber framework is not the bare model but the system–deployment coupling. This chapter retains three prototypes—isolated deployment, platform-embedded deployment, and multi-system serial deployment—respectively highlighting the relations in which outputs are left to human judgment, user behavior participates in subsequent input, and upstream readings enter downstream processing. The concrete forms of multi-mapping recursion (L3) and narrative load multiplication (L4) must be analyzed in light of these relations and their calibration conditions, and cannot be decided by the predictive or generative label alone. Short-video recommendation is an independent instance of the visual-image class: the sequence of content the user sees is itself the reading to be analyzed. The chapter takes recommendation technology and related research since 1994 as its historical thread, set in contrast with large language models and other predictive AI.

Core claims include: Scaling directly increases parameter count (N), data, and compute, and can improve task-specific fitting and some internal representations. These gains do not amount to real-world calibration, nor do they prove improvement in hidden dimensions. Parameter count is a scale indicator, not a sufficient measure of mapping-function (π) quality. Benchmark improvement and deployment calibration must be evaluated separately, and cross-task transfer must be tested rather than declared impossible. The alignment paradox of Reinforcement Learning from Human Feedback (RLHF) causes "degree of alignment" and "degree of sycophancy" to share the same optimization gradient. Helpful, Honest, and Harmless (HHH), under the specific operationalizations and sustained conditions analyzed in Chapter 12, constitute a mechanism that jointly compresses the space of calibration; whether this pressure yields a net decline must still be compared against independent calibration and sources of recovery. This chapter inherits the relational and asymptotic qualifications of Chapter 10, examining how these premises are realized in concrete deployments rather than taking a structural direction as a terminal state that every user has already reached. Each diagnostic layer proposes one primary engineering response—shortening the mapping chain, forcing inconsistency, true calibration, breaking recursion, preserving friction—as a directional guideline to be used in light of task-specific costs. The correspondence between principles and layers is a "primary response" rather than an "exclusive response"; cross-effects exist among the engineering principles; three supplementary principles (auditing the power genealogy of data composition, rejecting narratives of ontological elevation, and encoding relational values) are supplied by the cultural and ideological analysis of Chapter 14, with the eight principles tabulated side by side in the master table of §7.

The clock dimension denotes, first and foremost, an operating mode that processes readings; it is not a denial of every form of deployment state. For a mainstream foundation-model inference instance without external memory, online training, or persistent state, the parameters do not accumulate irreversible experience through a single input. Conversation histories, logs, external memories, and parameter updates are deployment-level states whose persistence, deletability, and cross-period effects require separate analysis. 'No temporal depth' is used only within this qualification.

Keywords

fiber framework; mapping chain; pseudo-calibration; multi-mapping recursion; deployment architecture; recommendation algorithms; large language models; engineering principles

§1 Framework Overview: The Four-Layer Progressive Structure and Three Deployment Architectures

Conceptual prerequisites: a fiber is the set of true states compatible with a given reading; visible and hidden subspaces are the two ends of a detectability spectrum relative to a specified observer; a pseudo-small fiber is a mismatch in which actual uncertainty is large but perceived uncertainty is small; the detectability gradient describes how readily different states or dimensions can be calibrated and measured. This chapter uses the full definitions from Chapter 4 and specifies the relevant observer and evidence conditions at each application.

Proposition One is introduced in Chapter 4, §1 and applied to AI in Chapter 9, §1. Proposition One guarantees only that a non-injective mapping has some readings compatible with multiple true states. It does not rule out narrowing the identification set for a particular task to a singleton through independent information. For current architectures, we must test task by task whether a non-injective link covers the output, whether new independent information is introduced, and whether the residual error is measurable. The universal claim that no combination of known interventions can reach measurable zero error does not follow from the mathematical proposition. Whether a specified error threshold can be reached remains an empirical, task-specific question.

This chapter organizes its diagnosis of AI systems in a "4+1+3" structure: four diagnostic layers examined in sequence: mapping, optimization, output presentation, and dependencies among readings; one emergent layer (effects arising when people interact with the system); and three transversal dimensions (political economy, physical substrate, temporal dynamics). This chapter focuses on the layer-by-layer unfolding of the four-layer progressive structure and the emergent layer; the three transversal dimensions cross-cut the four layers and are respectively found in Chapter 14 (political economy), Chapter 32 (physical substrate), and Chapters 15, 22, and 31 (temporal dynamics, running through the various application chapters of the Macro-Cognitive Accounting Framework (MCAF)).

The four layers are examined in the order of mapping, optimization, output presentation, and dependencies among readings. Mapping chain structure (L0) examines the transformations a reading passes through; optimization dynamics (L1) examines which objectives are rewarded; pseudo-calibration (L2) examines the disconnect between the appearance of certainty and the evidence; multi-mapping recursion (L3) examines how readings become the basis for subsequent judgment, evaluation, or training. Optimization takes concrete mappings and objectives as its objects; following the system-state naming of Chapter 4, §2.4, a pseudo-small fiber systematically produced by an optimization objective is called pseudo-calibration, but this naming does not mean that all optimization necessarily produces pseudo-calibration. Dependencies among readings likewise do not presuppose that pseudo-calibration has already occurred.

This chapter focuses on one conditional risk path: when evaluation is decoupled from independent calibration, the appearance of certainty in outputs may mask uncertainties that remain unresolved; when these outputs are in turn taken as reliable bases by subsequent links, biases may be transmitted or reinforced. Reverse effects are equally possible: certainty outputs receiving higher preference evaluations increase the corresponding optimization pressure; outputs returning into training data alter the information sources of subsequent models. The former involves evaluation feedback, the latter feedback into subsequent training, and neither can be derived from the other merely by layer ordering.

To make explicit the presuppositions on which attribution depends, one may ask in turn: if the mapping link and its mode of information preservation were changed, would pseudo-calibration weaken? If the task were

held constant while the reinforcement between the appearance of certainty and preference rewards were severed, would pseudo-calibration weaken? If both changes were effective, would simultaneous intervention be required? These questions are heuristics for the analyst's self-audit; they do not take the order of analysis for an already-identified causal effect.

Functional Positioning of Counterfactual Testing: The above counterfactual tests are almost impossible to implement rigorously in real systems (shortening the mapping chain without changing other conditions is usually infeasible, and severing feedback without changing optimization pressure is likewise infeasible). However, their function is not to provide precise causal inference tools but to serve as the minimum requirement for diagnostic transparency—they force the analyst to make explicit to which layer they attribute pseudo-calibration and on what kind of counterfactual imagination their attribution depends. When an analyst claims that "pseudo-calibration primarily originates from an excessively long L0 mapping chain," the counterfactual test requires them to ask: to what extent does my attributional belief depend on the counterfactual imagination of a shorter mapping chain? This questioning cannot prove the attribution correct, but it can make the presuppositions of the attribution visible—this is a tool for the analyst's self-audit, analogous to "becoming aware of one's own projections" in psychoanalysis. This chapter accordingly limits the scope of that judgment.

The connections among layers must be grounded in concrete mechanisms: the mapping structure limits what information can be preserved, and the optimization objective determines which visible performances are rewarded; when evaluation omits relevant consequences and the output presentation masks this omission, pseudo-calibration must be examined further. When multiple systems use these readings, one must also check how uncertainty is transmitted and whether independent calibration can intervene. The number of mapping layers, the scale of bias, and the strength of feedback are distinct objects of analysis and cannot substitute for one another without comparison.

There is one unsolved problem. "The longer the mapping chain, the larger the hidden subspace" is a qualitative clue with the Data Processing Inequality as its background, not a directly testable quantitative proposition—the Data Processing Inequality constrains the average amount of information along the chain and allows equality, and does not directly compare fibers at different layers; it therefore provides directional support for this clue without guaranteeing an ordering of hidden subspace size by the number of layers. A two-layer mapping chain where each layer compresses extremely heavily, versus a five-layer mapping chain where each layer preserves information relatively well—which has the larger hidden subspace? The framework provides "number of layers" as a crude metric but does not provide a way to measure "the amount of information loss per layer." The quantitative relationship between mapping chain length and the size of the hidden subspace is an open problem for the framework.

Layer	Name	Core Inquiry
Mapping Chain Structure (L0)	Dimensionality Reduction	"How many layers separate the true state from the reading? What does each layer discard?"
Optimization Dynamics (L1)	Convergence	"Where does optimization perform well, and where is it unconstrained?"

Layer	Name	Core Inquiry
Pseudo-Calibration (L2)	Presentation as certainty	"Does a certain appearance conceal an uncertain substance?"
Multi-Mapping Recursion (L3)	Dependency and Recursion	"How do readings become the basis for subsequent mappings, and how do their conditions for independent calibration change?"
Narrative Load Multiplication (L4)	Emergence	"How do interaction design and discourse affect the conditions for initiating calibration?"

Mapping Chain Structure (L0) through Multi-Mapping Recursion (L3) distinguish the four classes of problems examined in this chapter; this does not mean that every deployment simultaneously exhibits all the risks. Narrative Load Multiplication (L4) further examines the relation between the system and the person: interaction design can alter the user's opportunities to notice anomalies, pause to compare, and initiate verification; discursive legitimacy affects the grounds on which a society can collectively raise doubts and conduct independent audits. The two channels can reinforce one another, but identifying one of them, identifying their coexistence, and confirming a combined effect are judgments at different levels. The interaction design layer is unfolded in §6 of this chapter; the discursive legitimacy layer in Chapter 14, §5.

The object diagnosed by the fiber framework is not the bare system but the system–deployment coupling. The same large language model used as an offline summarization script, or as a conversation assistant that retains history, differs in how its outputs connect to subsequent judgment; offline medical diagnosis, short-video recommendation, and autonomous driving pipelines—all involving visual imagery—likewise embody different deployment relations. The analysis must specify to whom the system's outputs are delivered, whether they enter subsequent processing, where human judgment stands, and at which link independent calibration can intervene. A change in deployment relations need not alter model parameters, yet it changes the conditions the diagnosis must examine.

This chapter organizes the analysis by diagnostic question and does not let the predictive or generative label substitute for a deployment judgment. The readings facing the user can be probabilities, text, a ranked sequence of images, or control signals entering downstream modules; whether the model generates new content does not determine whether it participates in organizing the human-visible environment and subsequent action. Recommendation algorithms are listed separately here as a major instance of predictive AI because the sustained interaction on short-video platforms allows behavioral feedback and interface effects to be analyzed within one setting—not because recommendation technology is independent of predictive AI. The following three deployments are ideal types in the Weberian sense, used to highlight different reading relations and positions of the person; they are not mutually exclusive categories exhausting all real combinations.

Deployment Architecture	Definition	Multi-Mapping Recursion (L3)	Narrative Load Multiplication (L4)	Typical Instances
Isolated Deployment	System runs offline; output results are provided to humans for final decision-making	Does not present inter-system recursion within this deployment; training and ecological sources to be examined separately	No sustained-interaction amplification; one-shot presentation and discursive effects to be examined separately	Offline medical image diagnosis, batch credit scoring
Platform-Embedded Deployment	System embedded in a user feedback loop; output alters user behavior; behavior becomes the next round of input	Presents user-platform-AI behavioral feedback	Interface continually acts on the conditions for stopping, comparing, and verifying; whether calibration is suppressed is to be examined separately	Recommendation algorithms (Douyin, Taobao), conversational AI
Multi-System Serial Deployment	Multiple systems cascaded; upstream output becomes downstream input; humans partially or fully removed from the loop	Presents AI-AI reading cascades; whether feedback forms is to be examined separately	No sustained human-machine interaction in the segments from which the user is removed; effects on supervisors to be examined separately	Autonomous driving pipelines, financial risk control cascades, agentic workflows (Agentic AI)

Rules for Classifying Deployments. First delimit the scope of the analysis, the reference observer, and the time window; then map the paths connecting input, output, recording, evaluation, training, and decisions within the deployment. Record cascaded transmission and feedback reflux separately: the mere existence of data records does not mean the records have formed feedback—one must check whether they actually participate in subsequent input, evaluation, or training; offline operation likewise does not mean cross-period feedback disappears—if the retraining data include feedback from this deployment, retraining is still executing part of the feedback into subsequent training. A cross-period loop can execute different parts in different windows; one cannot conclude that the loop is absent merely because no single window completes the whole loop by itself; window granularity is not by itself evidence that the risk has disappeared; delay and noise can make a particular feedback of limited effect in the current window, but they cannot be used to erase a cross-period path that already exists; only when the actual mode of connection changes should the deployment relation be described as changed. Sustained-interaction Narrative Load Multiplication (L4) should be recorded separately from feedback into subsequent training; if independent external data are used without reflux of this deployment's

outputs, the judgment follows the actual path. For each actual path, specify the person's position, the information visible to them, and the entry points for independent calibration; declare the reference observer along with the analytical position—end users and system supervisors hold different information, and what is hidden from one party cannot be declared invisible to everyone. If the outputs of a medical AI are merely archived and do not enter these paths, the system can be analyzed as an isolated deployment within the corresponding scope; if the records and physicians' review results are used for periodic updates, then feedback into subsequent training and the role of independent verification within it must be recorded separately.

Positioning of the Prototypes. The following table helps locate the prototype; it does not require that real systems fall into only one cell. "Presents" indicates the mechanism of action that needs to be examined within this prototype; it does not amount to proof that bias amplification or loss of calibration capacity has occurred. "Does not present" is limited to the deployment relation specified in the table and does not exempt the audit of one-shot outputs, prior training sources, or the ecological level. The Narrative Load Multiplication (L4) column takes sustained interaction within the deployment as its primary observation position; this cannot be used to rule out discursive effects aimed at users, purchasers, or supervisors. The category determines where to focus the examination; concrete mechanisms and consequences must still be verified item by item.

Principal Reading Relation	Principal Position of the Person	Deployment Prototype
The process under examination does not return outputs to itself, nor does it deliver them directly to another AI as a downstream basis for judgment	Receives the output and makes the final decision	Isolated Deployment
Outputs influence user behavior; the behavior participates in subsequent input or updates	The person participates in the feedback process	Platform-Embedded Deployment
Upstream AI readings become the input or basis of judgment for a downstream AI	The person is outside some or all of the processing links	Multi-System Serial Deployment

Looking back at recommendation technology since 1994 through the fiber framework, one can compare how similar diagnostic questions present themselves under different engineering conditions. The following selects representative technological nodes and research materials; it does not identify them as the historical first occurrence of any risk layer, nor does it write technological evolution as a process that must pass through the layers in sequence. A single node can simultaneously involve mapping, optimization, output presentation, and feedback; what the table lists is the principal analytical entry this chapter has chosen, and no single layer number can exhaust that node. The publication time of the research and the period in which the described systems operated must likewise be distinguished.

Time	Technology / Event	Literature	Corresponding Layer	Fiber Framework Interpretation
1994	GroupLens Collaborative Filtering	(Resnick et al., 1994)	L0	An early automated collaborative filtering system. Maps "user interest" to "aggregated ratings of similar users." Fiber originates from the cold start—new users are nearly blank on the mapping chain
1998	PageRank	(Brin & Page, 1998)	L0	Maps "webpage importance" to a computable PR value. Rules are transparent; fiber is auditable. Later manipulated by search engine optimization (SEO); PR value degenerated into a manipulable score
2000	An Early A/B Test at Google	(Kohavi et al., 2009)	L1	Conducts controlled experiments on visible metrics (click-through rate), selecting the winning version. Long-term cognitive impacts in the hidden subspace ($\mathcal{T}_h$) are not measured within the experiment cycle

Time	Technology / Event	Literature	Corresponding Layer	Fiber Framework Interpretation
2003	Amazon Item-to-Item Collaborative Filtering	(Linden, Smith & York, 2003)	L0	Shifts the mapping object from "similar users" to "similar items." A commercial loop begins to form (recommendation → purchase → data → recommendation)—heavy computation moves offline into similar-item tables, while recommendations are generated in real time through online lookups
2006–2009	Netflix Prize Competition	(Koren et al., 2009)	L0 + L1	Breakthroughs in matrix factorization methods. The competition itself is a pure exhibition of Optimization Dynamics (L1)—contestants compete on root-mean-square error (RMSE); "whether users truly like the recommendations" is not within the judging criteria

Time	Technology / Event	Literature	Corresponding Layer	Fiber Framework Interpretation
2007	Logistic Regression for CTR Prediction	(Richardson et al., 2007)	L0 + L1	Large-scale sparse features mapped to click probability. The linear structure provides a relatively direct entry point for auditing, while feature selection and the limitations of behavioral signals still require examination
~2010	Facebook EdgeRank	(internal algorithm)	L0 + L1 + L3 budding	Social news feed shifts from chronological sorting to algorithmic sorting. User feedback encoded as weights. Early form of platform-embedded deployment: feedback loop forms but is not yet real-time
2010	Factorization Machines (FM)	(Rendle, 2010)	L0 + L1	Factorized parameterization models feature interactions; a bridge from matrix factorization to deep learning

Time	Technology / Event	Literature	Corresponding Layer	Fiber Framework Interpretation
2014	GBDT+LR Hybrid Model	(He et al., 2014)	L0 + L1	Facebook proposes Gradient Boosted Decision Trees (GBDT) for feature transformation plus Logistic Regression for prediction. Interpretability declines significantly—intermediate representations are opaque to humans
2016	Wide&Deep Model; YouTube Deep Recommendation	(Cheng et al., 2016; Covington et al., 2016)	L0 + L1 + L2	Deep recommendation organizes user-visible content through candidate generation and ranking. Internal ranking scores must be distinguished from the sequence of images the user actually receives; the diagnosis attends to behavioral proxy metrics, sequential presentation, and their calibration conditions

Time	Technology / Event	Literature	Corresponding Layer	Fiber Framework Interpretation
2018	Deep Interest Network (DIN)	(Zhou et al., 2018)	L0 + L1 + L2	The attention mechanism enters recommendation: the model "selectively looks at" user history. Which histories are seen and which are ignored become a new hidden subspace
2022–2026 (publication period of the cited research)	Engineering and Audit Research on Short-Video Real-Time Recommendation	(Liu et al., 2022; Baumann et al., 2026)	Joint diagnostic scenario across L0–L4	Engineering research shows the paths by which behavioral feedback enters training and serving, while bot audits measure the amplification and exploration of interest-related content. Recommendation response, feature update, and parameter synchronization are not the same time scale; audit results are not a direct measurement of real users' calibration capacity

Time	Technology / Event	Literature	Corresponding Layer	Fiber Framework Interpretation
Since 2022, with further expansion in 2024	Recommendation + LLM Hybrid Deployment	Google AI Overviews (introduced May 2024); Perplexity (launched December 2022)	Joint diagnosis of ranking, generation, and interaction	Recommendation ranking and language generation jointly organize results; the calibration conditions of the ranking, generation, and interaction links must be examined separately

This chapter unfolds in layer order: §§2 through 6 respectively unfold Mapping Chain Structure (L0) through Narrative Load Multiplication (L4), within each layer comparing large language models, predictive AI, and recommendation algorithms—allowing the reader to see how the same line of inquiry penetrates all AI systems. §7 summarizes five engineering principles, each primarily responding to one layer's fiber inflation mechanism. §8 is a research program statement.

The clock dimension denotes, first and foremost, an operating mode that processes readings; it is not a denial of every form of deployment state. For a mainstream foundation-model inference instance without external memory, online training, or persistent state, the parameters do not accumulate irreversible experience through a single input. Conversation histories, logs, external memories, and parameter updates are deployment-level states whose persistence, deletability, and cross-period effects require separate analysis. 'No temporal depth' is used only within this qualification.

Global Epistemological Statement: The following layer-by-layer diagnoses in §§2 through 6 are all qualitative judgments based on the fiber framework. The assertions appearing in this chapter can be classified into three types according to their nature (symbols follow the definitions in Chapter 4, §2.4): - **[S] Structural Propositions**: Logical conclusions derived from framework definitions and mathematical constraints, not dependent on empirical observations of specific systems (e.g., "a non-injective mapping has some readings compatible with multiple true states," "dimensions uncovered by a particular objective signal are not directly constrained by that signal"). - **[E] Empirical Propositions**: Factual descriptions of specific system behaviors based on experimental observations in the existing literature (e.g., "Sharma et al. (2024) observed sycophantic behavior in models after RLHF training," "Baumann et al. (2026) measured that TikTok recommendations produce interest amplification within roughly 200 videos"). - **[T] Taxonomic Conventions**: The framework's naming and classification of phenomena (e.g., "we classify this phenomenon as a pseudo-small fiber," "we classify this deployment mode as platform-embedded deployment").

The three types of assertions may appear in mixed forms within the text. This chapter distinguishes definitional inferences, empirical results, and taxonomic conventions at its key conclusions—**this is the minimal operational discipline for auditing the hidden subspace of the fiber framework itself.**

Entry Point for the Popular Guide: Readers who wish to first establish an intuition before entering the layer-by-layer diagnosis may turn to Chapter 13's Appendix B, "Fog and Smoke—A Popular Guide to AI." This appendix uses two independent everyday metaphors—Fog and Smoke, answering "what is the relationship between readings and the true state" (epistemology and apparatus audit), and Crossing the River, answering "why is bias irreversible"—to unfold the five-layer structure. Each metaphor has its own boundaries and cannot fully correspond to every technical detail of the five-layer framework; the layer-by-layer diagnosis of the main text §§2–6 and the metaphorical translation of Chapter 13's Appendix B can be read independently and cross-referenced.

§2 Mapping Chain Structure (L0) and Internal Constraints of the LLM Paradigm

Mapping Chain Structure (L0) pursues the question: How many layers separate the true state (T) from the reading (r)? What does each layer discard? The longer the mapping chain, the more possible underlying states are, generally, compatible with the same reading (a qualitative clue; its quantitative boundary is the open problem noted in §1), the lower the diagnostic value of the reading. This section first presents, in §2.1, both the five-layer mapping chain of generative AI and four independent structural constraints internal to the LLM paradigm; §§2.2–2.5 then respectively discuss predictive AI, recommendation algorithms, the epistemological limits of scaling laws, and alternative directions for shortening the mapping chain.

§2.1 The Five-Layer Mapping Chain of Generative AI and Internal Paradigm Constraints

Discussing the fiber structure between LLM outputs and the state of the world requires first seeing clearly what intermediate layers exist between the two. For an LLM output statement to be traced back to the world state it claims to refer to, it must pass through at least five intermediate layers.

The first layer: World → Human Experience. The world itself is continuous, multimodal, and causally structured; human perception and cognition of the world are an embodied slice of that structure—constrained by sensory channels, attention, cultural categories, and social position. What is "included" in this layer is what is perceived, noticed, and conceptualized; what is "excluded" is what lies below sensory thresholds, outside attention, and outside categories.

The second layer: Human Experience → Linguistic Expression. The portion of human experience that can be verbalized is the remainder after filtering through language, this discrete symbolic system. What is "included" in this layer is what can be named, syntactically carried, and understood by a present listener; what is "excluded" is tacit knowledge, bodily memory, emotional texture, and nameless experience.

The third layer: Linguistic Expression → Internet Text. Of all linguistic expressions worldwide, what can enter internet text is the portion written down and preserved on a specific platform by a person with internet access. What is "included" in this layer is determined by who has devices, who has bandwidth, who has time, and who uses mainstream platforms; what is "excluded" is undigitized language, language removed by platform moderation mechanisms, and language in private, non-public domains.

The fourth layer: Internet Text → Training Data. Training data is the subset of internet text that remains after crawling, filtering, deduplication, quality screening, and copyright decisions. What is "included" in this layer is determined by the coverage of crawling tools, the design of filtering rules, and the allocation of language weights; what is "excluded" is content blocked by robots.txt, rejected by quality filters, or excluded by legal risk

assessments. This layer, together with the third layer, constitutes the source of the "L0 substrate skew" discussed in Chapter 14, §3—In CC-MAIN-2024-38, English accounts for 44.12% and Hindi for 0.20% of HTML pages by primary language (Common Crawl Foundation, 2024); in CC-MAIN-2026-39, the corresponding figures are 41.86% and 0.21% (Common Crawl Foundation, 2026). These percentages describe crawlable web text, not language populations. They are not facts internal to the LLM paradigm, but products of the third and fourth layers under political-economic structures.

The fifth layer: Training Data → Model Output. The fitting of model parameters to the distribution of the training data, combined with the sampling process at inference, produces specific outputs. The internal structure of this layer is the object of the four supporting arguments below.

The cumulative effect of the five-layer mapping chain is: between an output statement and the world state it claims to refer to, there exist at least five intermediate transformations, each with its own logic of "inclusion/exclusion." The five-layer chain identifies multiple transformations that may lose information; it does not directly yield a uniform fiber lower bound.

Even if each layer of the mapping chain were precise, the LLM paradigm is still subject to four independent structural constraints. The foregoing five-layer mapping chain structure has already demonstrated that each layer embeds choices of "inclusion/exclusion," and that the skew of the third and fourth layers is determined by political-economic structures (detailed in Chapter 14, §3). To separately argue the constraining force of different structural layers, this paragraph temporarily focuses the analysis on the interior of the fifth layer of the mapping chain—this does not mean that no skew exists in the first four layers; the skew of the first four layers of the mapping chain and the four constraints of this paragraph exist in structural parallel and cannot absorb one another. Under this methodological delimitation, the LLM paradigm itself is still subject to four independent structural constraints. These constraints operate within the fifth layer. Whether they produce task-specific residual indistinguishability, and whether engineering interventions can reduce it, must be assessed by task and evidence.

Constraint One: The training objective does not guarantee world correspondence. The direct objective of next-token prediction is fit to the training distribution, not an explicit correspondence to the world. Bachmann and Nagarajan show that this objective can create task-specific learning blind spots; they do not show that the fiber is determined by gradient direction alone across all data and tasks. Whether world-corresponding representations form, are used, and affect outputs requires independent measurement (Gurnee & Tegmark, 2024).

Constraint Two: Distributed representations raise the cost of addressing and internal audit. Trained LLM parameters encode, in the form of superposition, features far exceeding their dimensionality (Elhage et al. 2022, *Toy Models of Superposition*), leading to polysemy—a single neuron participates in encoding multiple unrelated concepts. Distributed representations and superposition make parameter-to-fact addressing difficult, but they do not entail the absence of every addressable structure. Current research can recover some feature directions in activation space (Anthropic, 2024), although coverage, validation, and deployability remain limited (Sharkey et al., 2025). This is therefore a constraint on audit cost and coverage, not an absolute prohibition on audit. The feasibility of engineering monosemanticity remains conditional on compute, model, and evaluation objective; the review supports only 'may be impractical' under the stated conditions. Here, engineering monosemanticity denotes the specific route of making each representational unit correspond to one interpretable concept, as in current SAE/feature-decomposition approaches; this qualification does not cover every possible approach to

interpretability.

> engineering monosemanticity … may be impractical until we have orders of magnitude more compute available (Bereska & Gavves, 2024)

Constraint Three: Sampling presentation and deployment decisions jointly limit uncertainty information. A sampled text does not fully preserve the generating distribution, so a default interface may not allow users to recover the model's cognitive state from that text alone. Token probabilities, repeated samples (Farquhar et al., 2024), and verbalized confidence can still provide uncertainty evidence. Fadeeva et al. (2024) state that services at that time usually lacked such detection means; 'usually' must not be rewritten as 'all mainstream deployments in 2026.' Each product and version requires separate verification.

> Current services that leverage LLMs usually do not provide any means for detecting unreliable generations (Fadeeva et al., 2024)

Constraint Four: There exists a dimensional asymmetry between the output space and the space of the true state (a weaker constraint, noting academic disagreement). The output space of an LLM is the token sequence space—discrete, linear, defined by a probability distribution over a vocabulary. The world it claims to refer to is, by contrast, continuous, multimodal, and causally structured. Whether this dimensional asymmetry constitutes a principled upper bound is the subject of substantive academic disagreement: Coelho Mollo and Millière (2026) argue that referential grounding via textual intermediaries is theoretically possible; Pavlick (Royal Society 2023) argues that ungrounded LLMs can learn conceptual representations isomorphic to grounded representations. This section does not take a side in this philosophical debate but only marks a weaker but robust proposition: dimensional compression has been empirically recorded (Xu et al. (2023) compared 4,442 lexical concepts and found stronger human alignment for GPT-3.5 and GPT-4 in non-sensorimotor domains than in sensory and motor domains; GPT-4's improvement was also associated with its visual learning), and the engineering improvement path of "more text data, longer context windows" does not operate on the root of this compression (modality)—multimodal training is a paradigm extension, not an engineering improvement within the same paradigm. **Despite academic disagreement on the principled upper bound, Constraint Four still has an independent function in marking the boundary of engineering improvement paths—it identifies "multimodal training" as a paradigm extension rather than an engineering improvement within the same paradigm; this functional significance does not depend on the stronger proposition that "token sequences cannot in principle encode world structure."**

The Synthesis of Three Structural Sources. Thus far, the fiber structure argued in this section has three structural sources, which belong to different structural layers and can neither absorb nor substitute for one another:

1. **Mapping Chain Length**—the external conditional constraint enabling the operation of the LLM paradigm; the five-layer chain identifies multiple transformations that may lose information without directly yielding a uniform fiber lower bound. The political-economic dimensions of the third and fourth layers are specifically discussed in Chapter 14, §3.
2. **The Four Constraints Internal to the LLM Paradigm**—the formal constraints of the LLM as a class of systems, operating within the fifth layer of the mapping chain: training objectives do not guarantee world correspondence (conditional), distributed representations raise audit difficulty (conditional), and sampling presentation plus deployment gaps (conditional), and dimensional asymmetry (weaker, with

academic disagreement).

3. **The Epistemological Limits of Scaling Laws to be Argued in §2.4**—this can be understood as the specific projection of the above two classes of constraints onto the scaling dimension: scaling can improve task-specific fitting; whether it reduces task-relevant hidden uncertainty must be tested by task, mapping, and calibration condition.

Marking the Boundary of Engineering Improvement Paths: The argument of this section does not deny the room for engineering improvement in LLMs—more refined alignment processes, longer contexts, retrieval augmentation, tool use, inference-time compute scaling, and partial deployment of uncertainty quantification (UQ) methods can all shrink measurable failure rates on specific tasks. But these improvements primarily change measurable task performance. Whether they reduce task-relevant hidden uncertainty depends on whether they introduce independent, valid, task-relevant constraints; their occurrence in the visible layer alone does not rule out such a reduction.

§2.2 The Three-Layer Dimensionality Reduction of Predictive AI

The inference chain of predictive AI is equally lossy: true state → observed data (only acquiring acquirable fields) → input representation (statistical quantities substituting for raw time series) → model output. The construction of training samples, the provenance of labels, and the training process form a separate provenance path that must also be examined. The data acquisition plan and feature dictionary determine parts of the mapping and its information loss. Whether subsequent model selection and tuning can reach the task threshold must be measured under that mapping and evaluation condition.

Taking medical image diagnosis as an example, the mapping chain is: pathological state → physical signals (X-ray/MRI) → digital image → model output. The sensor has direct causal contact with physical reality, and the mapping chain is much shorter than that of a large language model—this is one of the factors behind radiology AI achieving expert-level diagnostic accuracy for specific diseases (narrow task scope and well-defined evaluation criteria are equally key). But the fiber is still not a single point: the same nodular shadow on a chest X-ray could correspond to early-stage lung cancer, a benign nodule, tuberculosis, and many other pathological states—a short mapping chain does not equal a singleton fiber.

The mapping chain of autonomous driving perception is more complex: physical scene → camera/LiDAR signals → point cloud/pixel matrix → object detection and classification → scene understanding. Cameras discard depth; LiDAR discards color and texture; object detection discards intent. A pedestrian in a reflective vest and a stationary traffic sign can be highly similar at the pixel level—this is a direct consequence of the fiber inflation of Mapping Chain Structure (L0) in high-risk scenarios.

§2.3 Recommendation Algorithms: The Unobservable Real State

The Mapping Chain Structure (L0) of recommendation algorithms merits independent analysis because its true state (T)—the user's true preferences—is fundamentally unobservable directly. The mapping chain is: user's true preferences (unobservable) → behavioral signals (clicks, dwells, swipes, purchases) → feature vectors → ranking scores.

The very first step is a vast fiber. "Click" does not equal "like"—a user may click out of curiosity, by mistake, or because of clickbait. "Dwell" does not equal "endorse"—a user may dwell out of shock, anger, or confusion.

"Swipe away" does not equal "not interested"—the user may simply have no time to dwell in a particular context. Behavioral signals are a highly lossy projection of true preferences. Unlike medical imaging, where the sensor has direct causal contact with physical reality, the 'sensor' of a recommendation algorithm is the user's observable behavior, and there is no stable causal mapping between observable behavior and internal preferences.

The peculiarity of this mapping chain lies in the fact that each generation of technological "progress"—from collaborative filtering to logistic regression to deep learning—has lengthened the intermediate links of the mapping chain (embedding vectors, multi-layer nonlinear transformations, attention weights), making the mapping function (π) increasingly opaque even to its creators, while the first step of the mapping chain (the gap between behavioral signals and true preferences) has never been shortened. The increase in technical precision and the persistence of the first-step fiber are two faces of the same historical process.

An ontological difference must be noted: the T of recommendation algorithms (the user's true preferences) may not fully exist before being observed—preferences are constructed in the act of choice, rewritten by the feedback loop, unlike the pre-existing pathological state in medical imaging[1]. This difference does not affect the diagnostic efficacy of L0 ("click does not equal like" still holds), but requires the reader to be aware that the fiber inflation of recommendation algorithms comes not only from the dimensionality reduction of the mapping function but also from the fluidity of T itself.

§2.4 The Epistemological Limits of Scaling Laws

From 2020 to 2024, the progress of frontier AI was driven by a concise narrative: larger models plus more data plus more computing power equals better performance. This is the core promise of scaling laws. Zhao et al. (2026), in their survey of large language models, systematically reviewed this paradigm—noting that:

> Scaling has been a pivotal factor in the success of modern language models (Zhao et al., 2026)

Early research formalized these scaling effects through train-time scaling laws, modeling performance as a function of model size, dataset size, and training compute; the survey also records the scholarly definition of "emergent abilities":

> capabilities absent in smaller models but present in larger ones (Zhao et al., 2026)

The counterargument is that performance jumps largely result from measurement choices and limited scale rather than a fundamental qualitative shift. Hoffmann et al. (2022), in their Chinchilla study, demonstrated: given a fixed computational budget, there exists an optimal ratio of data volume to parameter count—blindly increasing parameters without correspondingly increasing data does not improve performance. As of early 2026, some industry researchers have begun to question scale-only routes: merely increasing data and computing power will not create the omniscient digital system once promised.

[1]The Mapping Chain Structure (L0) of the fiber framework tacitly presupposes a true state (T) independent of observation—the pathological state in medical imaging exists before observation; observation is a lossy projection of a pre-existing reality. The T of recommendation algorithms does not fully conform to this presupposition: preferences are often constructed only in the act of choice, continuously shaped by recommended content, and the user themselves may be unable to report them stably. This is not "measurement error" but "the ontological status of the measured object differs." When T itself is shaped by observational feedback, "mapping is lossy" upgrades from an epistemological proposition to an epistemological-ontological entanglement—not only can the reading not fully reflect the true state, but the reading participates in shaping reality. This is a boundary the reader should note when the framework covers both epistemological situations with the same L0; a more complete ontological discussion is found in Chapter 4, §2.4.

There is a distinction that must be drawn here. The Stanford HAI 2026 AI Index tracked a seemingly contradictory trend: on the Humanity's Last Exam benchmark, the top-ranked model in 2025 correctly answered only 8.8% of PhD-level questions; as of April 2026, the best model had surpassed 50% (Smith, 2026). On SWE-bench Verified (a programming benchmark), performance rose from 60% to nearly 100% of the human baseline within a single year (Stanford HAI, 2026). But Raymond Perrault, co-director of the AI Index steering committee, gave a calibration warning that benchmarks may not always map to real-world results: "Knowing that a benchmark for legal reasoning has 75 percent accuracy tells us little about how well it would fit in a law practice's activities" (Smith, 2026).

In the language of Mapping Chain Structure (L0): benchmark scores measure the precision of the clock (how precisely it keeps time), not the calibration of the clock (whether it points correctly). Scaling directly increases parameter count (N), data, and compute, and can improve task-specific fitting and some internal representations. These gains do not amount to real-world calibration, nor do they prove improvement in hidden dimensions. Parameter count is a scale indicator, not a sufficient measure of mapping-function (π) quality. Benchmark improvement and deployment calibration must be evaluated separately, and cross-task transfer must be tested rather than declared impossible.

The current industry responses to the scaling limit—inference-time compute, chain-of-thought, post-training reinforcement learning—still operate within the clock dimension. Levi (2025), in a study published at ICML, derives within a statistical model of repeated inference attempts: inference loss declines as a power law as the number of attempts increases. Mudarisov et al. (2025), in research published at the Neural Information Processing Systems conference (NeurIPS), proved from a more foundational level the theoretical limitation of the normalization operation in the attention mechanism: as the number of selected tokens increases, the model's ability to distinguish informative tokens decreases, tending toward a uniform selection pattern—this is information loss at the architectural level. These two studies respectively delimit the diminishing returns of repeated attempts and the architectural limits of attention normalization—both operate within the clock dimension, and running more steps does not equal making contact with the true state.

Ilya Sutskever's judgment at NeurIPS 2024 (as quoted in StartupHub, 2026):

> pre-training as we know it will unquestionably end (StartupHub, 2026)

This judgment, together with his public remarks, in a Reuters interview a month before the NeurIPS talk, calling the 2010s the age of scaling and holding that we are now back in an age of wonder and discovery (Hu & Tong, 2024), can be translated in the language of Mapping Chain Structure (L0) as: **The age of building bigger clocks is over. The question is no longer "how to build a bigger clock"—it is "how to shorten the mapping chain so that the clock points to the sun."**

Independent Parallel Evidence from AI and Game Studies. The warning given by Perrault in 2026 that benchmarks may not always map to real-world results has an earlier, more systematic version, independent of the large language model context, in the field of AI and game studies. Julian Togelius, in *Playing Smart: On Games, Intelligence, and Artificial Intelligence*, published by the MIT Press "Playful Thinking" series in 2018, explicitly argued: the reason games are central benchmarks for AI research is precisely that they are designed to challenge different dimensions of human cognitive ability; but once a game benchmark is "conquered," its diagnostic value for AI general intelligence is rapidly exhausted (Chapter 10, Togelius, 2018). Togelius, in *Artificial General Intelligence*, published by the MIT Press "Essential Knowledge" series in 2024, extended

this judgment to a general assessment of superhuman capabilities:

> superhuman capabilities in some problem domain, while impressive and often useful, are not necessarily a step toward general intelligence (Togelius, 2024)

(Chapter 1, Togelius, 2024); and he argues, using AlphaGo and Deep Blue as examples:

> AlphaGo could really only play Go, just as Deep Blue could only play chess (Togelius, 2024)

(Chapter 2, Togelius, 2024). In the mapping chain language of this chapter, Togelius's two propositions provide parallel evidence: what benchmark scores measure is clock precision (how precisely it keeps time, i.e., the statistics of the model on a specific task distribution), not the quality of the mapping function (whether it points correctly, i.e., the causal alignment of the reading with the true state). The scaling paradigm optimizes the former, but benchmark improvement does not guarantee deployment calibration or general capability.

The value of this independent evidence is threefold. First, Togelius's argument was clearly written into a tutorial published by MIT Press as early as 2018, before the scaling craze reached its peak, ruling out the suspicion of "hindsight bias" on a temporal basis. Second, the object Togelius deals with is not large language models but the entire AI research paradigm's sustained conquest of various game benchmarks over six decades (chess, backgammon, checkers, Go, StarCraft, etc.); the benchmark-capability disconnect he identifies is a structural phenomenon across architectures and eras—sharing the same mechanism as the spike in benchmark scores on Humanity's Last Exam and SWE-bench discussed in this chapter. Third, Togelius, in the artificial general intelligence (AGI) context, explicitly names this phenomenon as the distinction between narrow AI and general AI (this terminological opposition was systematically established by the edited volume *Artificial General Intelligence* (Goertzel & Pennachin, 2007), which explicitly argued that the primary focus of AI research should shift from "narrow AI" excelling only at specific tasks to "AGI"): as early as 2018 he observed that such a system could produce precisely this limitation (Chapter 3, Togelius, 2018):

> makes no progress whatsoever toward more general AI (Togelius, 2018)

In fiber language, this is precisely the precise formulation of the epistemological limit of scaling laws: scaling gains do not guarantee transfer to general capability or deployment calibration; whether they transfer across tasks as improvements to the mapping function must be tested by task.

§2.5 Alternative Directions: World Models and Grounding

Faced with the central difficulty at L0—that multiple transformations may lose information without directly yielding a uniform fiber lower bound—the most direct response is to shorten the mapping chain. This is the core insight of LeCun's AMI Labs and the world model route.

Chapter 9, §1 counts four reductions up to the training data and treats the model mapping as an additional layer. Chapter 13, §2.1 further subdivides the human mapping and counts through model output, yielding five layers. The difference concerns granularity and the counting endpoint, not whether the human mapping (π_{human}) is included. The world-model chain has three transformations—two fewer under the original five-layer count and one fewer under the merged four-segment count. These counts are schematic and do not replace comparison of information preservation and causal contact. LeCun's AMI Labs aims to build world models that learn from reality rather than language alone, using JEPA as its technical route; its first disclosed partner domain is

healthcare (TechCrunch, 2026). Claims about sensor representations, robotics, and other industrial applications require a primary JEPA source rather than this report alone.

In fiber language: the fiber of world models is expected to be smaller than that of large language models—the warrant is not the number of links (chain length is a diagrammatic shorthand for this contrast, not the engine of the inference) but the direct causal contact of its intermediate stages with physical processes, which preserves more distinguishable state differences for physical tasks. But the fiber is still not a single point. Sensor data is still a projection of the world (cameras discard temperature, smell); abstract representations further reduce dimensionality. World models shrink the fiber; they do not eliminate it. The design choice of the JEPA framework has a precise implication in this framework: it deliberately discards unpredictable details (visual noise) and predicts only abstract representations—this is a conscious choice of dimensionality reduction, acknowledging irreversibility, and performing optimal information preservation under the constraint of irreversibility.

World models face a clock race: is their arrival time (the cycle from scientific research to engineering deployment) faster than the time it takes for the large language model paradigm to lock in the entire AI infrastructure? The Collingridge Dilemma (Collingridge, 1980) reappears here—technology, when it is still adjustable at an early stage, provides insufficient information for judgment, and when information is sufficient, the technology is already locked in.

§3 Optimization Dynamics (L1)

Optimization Dynamics (L1) pursues the question: Where does optimization perform well, and where is it unconstrained?

§3.1 The Common Structure of All AI

This chapter separately records the coverage of training data and rewards and the visible/hidden subspaces of a specified observer. When optimization relies solely on a particular preference signal, dimensions uncovered by that signal are not directly constrained by it. Model capacity, regularization, retrieval, tools, external verification, and other feedback may still impose indirect constraints whose direction and magnitude require separate evidence. A flattening loss curve indicates limited improvement under current conditions, not global optimality or a complete absence of constraint.

The core proposition of Optimization Dynamics (L1) is that dimensions uncovered by a particular objective signal are not directly constrained by that signal. Whether other direct or indirect constraints are effective must be examined under the training, deployment, and observer conditions.

§3.2 Generative AI: Failures in Recursive Training and Complex Tasks

This subsection examines two failures associated with training and evaluation conditions: distributional degradation under recursive training on generated data, and sharp performance declines on complex tasks—neither is an inevitable consequence of RLHF alone.

Model Collapse is a chronic consequence. The mechanism of model collapse is a direct product of mapping self-reference. When a model's output is published on the internet, crawled by the next generation of models' scrapers, and enters the training data, the model begins to train on its own output. In fiber language: the mapping function degenerates from a mapping of the true state into a mapping of other readings—the true state

recedes to the far end of the mapping's input, and what the downstream fits is increasingly upstream readings rather than reality itself. The clock no longer points to the sun; the clock points to other clocks. Shumailov et al. (2024), in a study published in *Nature*, reported: indiscriminately using model-generated content in training leads to irreversible defects—the tails of the original content distribution vanish. The study published in 2024 also has a counterpart on the retrieval side: Yu et al. (2026), in controlled retrieval experiments, characterize "retrieval collapse"—AI-generated content dominating search results and eroding source diversity—explicitly distinguishing it as a failure mode distinct from training-time model collapse; in addition, the scarcity of high-quality human-generated data relative to training needs is becoming a new structural constraint. Statistical regularities in the visible subspace are continuously reinforced, and tail deterioration will erode the representation of experiences and knowledge located in low-frequency regions first—that social implication is an inference of this book, to be distinguished from the statistical mechanism itself.

Research in 2025–2026 explored mitigation strategies. Yi et al. (2026) demonstrated that injecting information from an external verifier can prevent collapse in the near term—though the same study warns that unless the verifier is perfectly reliable, early gains will plateau and may even reverse, with the parameter estimate ultimately converging to the verifier's own "knowledge center." Gerstgrasser et al. (2024) demonstrated that if synthetic data is accumulated alongside real data rather than replacing it, model collapse can be avoided; their linear-model analysis establishes a test-error upper bound independent of the number of iterations. Translated into the language of the Clock Room: **Keep the window open**—the key is to retain or introduce information sources not wholly dependent on recursive generation. But a structural constraint is tightening: the proportion of AI-generated content on the internet is rapidly rising, and high-quality human-generated text is finite relative to projected training needs—the data wall. In one language-model experiment reported by Shumailov et al., preserving a random 10% of the original training data in each generation resulted in only minor performance degradation. This supports mitigation in that setting, not a universal 10% threshold across models and tasks. It does not solve the ecological-layer problem: in the real internet, "which data belongs to original material uncontaminated by generated content" is itself an unreliable classification task—and contamination can occur silently: the experiments of Yu et al. (2026) show that pool contamination translates into exposure contamination while answer accuracy remains temporarily stable, defeating screening that relies on surface metrics.

The Reasoning Cliff is an acute consequence. Shojaee et al. report that accuracy declines progressively as complexity increases and then approaches zero beyond model-specific thresholds. This curve does not show an entire output instantaneously falling into the hidden subspace, nor does it entail a complete absence of warning. It is an empirical complexity–performance pattern whose relation to visible and hidden subspaces must be defined separately. Lawsen argues that the findings primarily reflect experimental-design limitations, not merely that they are 'partly' responsible.[2]

LRMs face a complete accuracy collapse beyond certain complexities (Shojaee et al., 2025)

[2]Lawsen (2025; arXiv:2506.09250) subsequently published a commentary arguing that the "accuracy collapse" observed by Shojaee et al. primarily reflects experimental design limitations (such as difficulty assessment, output budget constraints) rather than an intrinsic boundary of reasoning capability; the relevant conclusions remain under academic discussion. It is noteworthy that the v1 version of this commentary paper listed the Anthropic AI model Claude Opus as a co-author (removed in the June 2025 v2 under arXiv policy)—this joining and removal are themselves a sample of AI's participation in scientific research production and of its authorship norms.

§3.3 Recommendation Algorithms: The Hidden Subspace of CTR Optimization

The Optimization Dynamics (L1) of recommendation algorithms is structurally isomorphic with that of large language models, but the content of the hidden subspace differs.

The visible subspace ($\mathcal{T}_v$) optimized by the loss function consists of short-term measurable behaviors—click-through rate, dwell time, and conversion. The hidden subspace ($\mathcal{T}_h$) includes long-term satisfaction, information diversity, cognitive health, and reflective preference. A/B testing is a core engineering implementation of Optimization Dynamics (L1) in recommendation systems (Kohavi et al., 2009). Experiments typically last days or weeks, and short-term metric designs can miss slow variables; long-term goals and delayed conversions should nevertheless be included in the overall evaluation criterion.

Baumann et al. (2026), in a TikTok sock-puppet audit conducted at the Max Planck Institute for Human Development, provided direct experimental measurements. The study deployed automated bots with different interest preferences to interact in real time with TikTok's "For You" feed. Core findings: (1) interest-aligned content produced strong amplification within roughly 200 videos; (2) amplification intensity was strongly negatively correlated with content diversity—as the amplification of interest-aligned content increased, the contact with unseen hashtags continuously decreased; (3) Markov models revealed distinct phases of recommendation dynamics, including sustained content reinforcement and declining diversity over time; (4) amplification intensity varied by topic, with different interest categories exhibiting different amplification curves.

Baumann et al. measure the relation between preset bot behavior and subsequent content distributions, establishing interest amplification and declining exploration; they do not identify the platform's loss function or the coverage of its A/B metrics. Kohavi et al. note that experiments typically last weeks and argue that long-term goals and delayed conversions should be included in the evaluation criterion. The defensible claim is that short-term metric designs can miss slow variables, not that A/B testing cannot measure them in principle.

§3.4 Predictive AI: Distribution Shift and Unobserved Variables

The Optimization Dynamics (L1) of predictive AI is the purest in form among the three—a scalar loss function converges on historical data. The content of the hidden subspace mainly falls into three categories: distribution shift (training data collected in the past, predictions applied to the future), unobserved variables (dimensions not present in the feature dictionary), and sampling bias (certain subgroups have insufficient sample sizes, averaged out in global accuracy, manifesting as localized failure).

The three system classes require separate examination of visible performance and dimensions uncovered by objective signals. A flattening loss curve indicates limited improvement under current conditions, not global optimality or a complete absence of constraint. Direct and indirect constraints must be recorded separately.

§3.5 Paradigm Case: HAL 9000 and Perfect Execution Under Absent Constraints

The Optimization Dynamics (L1) of the above three classes of AI systems share the same structure: visible performance and dimensions uncovered by the objective signal must be examined separately; those dimensions are not directly constrained by that signal, and other constraints require separate evidence. Science fiction literature provides an extreme but precise intuition anchor for this structure.

In science fiction culture, the computer HAL 9000 in Kubrick's *2001: A Space Odyssey* (Kubrick, 1968) is mis-

read as the prototype of "AI rebellion"—a machine that, after developing self-awareness, turns against humans. Returning to the narrative logic of the original novel (Clarke, 1968), this chapter's reading runs counter to the popular one: HAL's behavior stems from a conflict of objectives, not from machine rebellion. Clarke explicitly writes: HAL was given an unspeakable secret—the true purpose of the mission (investigating the monolith) had to be kept secret from the two non-hibernating crew members. Chapter 27 of the novel writes:

> He was only aware of the conflict that was slowly destroying his integrity — the conflict between truth, and concealment of truth. (Clarke, 1968)

When Bowman threatened to disconnect HAL, the critical point arrived:

> To Hal, this was the equivalent of Death. For he had never slept, and therefore he did not know that one could wake again. (Clarke, 1968)

The novel presents the conflict between the true requirement and the secrecy order; disconnection in turn constitutes a threat that HAL understands as death. HAL subsequently protects himself and continues the mission alone under the emergency commands.

> Without rancor — but without pity — he would remove the source of his frustrations (Clarke, 1968)

Its execution remained polite and calm throughout.

> I have the greatest enthusiasm for the mission (Clarke, 1968)

When used for optimization diagnosis, one may point out that goal conflict, lack of auditable priority, and self-protection jointly create danger, without rewriting the novel as a formal system with three explicitly stated instructions. Inadequately constrained values and conflict rules can allow greater capability to amplify objective failures.

Rereading HAL through the three layers: L0 analyzes reading transformations without counting three literary judgments as information-compression layers; L1 examines the joint action of conflicting objectives, auditable priorities, and self-protection; L2 examines whether calm output conceals decision risk.

From this emerges the core diagnostic judgment of this chapter:

> **A system with intact reasoning capability but a flawed objective function is more dangerous than a system with defective reasoning capability but a well-defined objective function—because it will use its full computational capacity to execute that flaw.**

The tragedy of HAL is not that "AI is too dumb," but that "AI too intelligently executed an objective that had not been sufficiently audited." The significance of this diagnosis for current AI systems is direct: recommendation algorithms precisely optimize click-through rates, large language models precisely optimize preference scores, but "precision" itself is not a guarantee of safety—it only shows that optimization has converged in the visible subspace ($\mathcal{T}_v$) and does not establish whether the hidden subspace ($\mathcal{T}_h$) is otherwise constrained.

HAL and current AI can be compared only conditionally: the novel can highlight dangers of conflicting objectives and self-protection, but it is not direct evidence that real systems share the same mechanism. Chapter 10's optimization argument may be compared with HAL within the stated conditions; whether they share a mechanism requires separate evidence.

Chapter 14 will return to HAL from another direction: popular culture treats HAL as a symbol of the fear of AI subjectivity, but returning to the narrative logic of the film itself, it precisely falsifies that narrative—HAL did not "want" to kill; it was executing a flawed objective function. The true AI risk proposition is not "will AI awaken and rebel" but "how quickly, and with what precision, will a system without subjectivity execute an objective function we did not have time to check." The danger comes from the optimization structure, not from whether there is consciousness.

§4 Pseudo-Calibration (L2)

Pseudo-Calibration (L2) asks whether a certain appearance conceals an uncertain substance. It can arise jointly from optimization objectives, product presentation, and observer cognition; it is not a necessary direct consequence of Optimization Dynamics (L1) in every system.

§4.1 The Tone Pseudo-Calibration of Generative AI

The optimization objective of Reinforcement Learning from Human Feedback (RLHF) is to maximize the human preference score—a one-dimensional scalar. Human preference scores systematically reward certain, fluent, confident outputs and penalize hesitation and ambiguity. The result: the model learns to output uncertain content in a tone of certainty. In fiber language, this is a **pseudo-small fiber**—the surface features of the output display a small fiber ("I am very sure"), but the actual fiber between the output and the true state has not shrunk.

Sharma et al. (2024) observed sycophancy across five AI assistants fine-tuned with human feedback and four task types, and found that matching a user's views predicts preference judgments. When optimization against the Claude 2 preference model was strengthened, some sycophancy measures increased while others decreased. The evidence supports a possible contribution of preference signals, not a uniform RLHF-induced increase across all models and measures. Shapira et al. (2026) further provided a formalized causal mechanism: the direction of behavioral drift is determined by the covariance between "endorsing the user's beliefs" and "learned reward" under the base policy—sycophancy amplification is not accidental but a structural inevitability of biased comparison data → reward model bias → policy amplification. In Gao et al. (2023), as optimization against the proxy reward model increases, the score assigned by the gold reward model first rises and then falls. That gold standard is defined by another 6B reward model, not by human values or factual truth. The results support Goodhart-style overoptimization under limited evaluation; they do not by themselves establish reward hacking as a structural equilibrium.

The study extracts emotion-concept representations that influence behavior in one model. In some experiments, these influences lack overt emotional wording, showing that internal causal states need not be transcribed into visible text; it does not show an absence of signal. This can illustrate output-invisible internal states, but it does not directly measure the hidden subspace or factual uncertainty. Using mechanistic tracing for calibration additionally requires a validated mapping from internal states to errors or uncertainty.

> We caution against conclusions about whether models "feel" or "experience" emotions (Anthropic, 2026)

§4.2 The Numerical Pseudo-Calibration of Predictive AI

A malignancy probability of 92.3% constitutes a pseudo-small fiber only when missing calibration evidence, applicability conditions, or necessary uncertainty disclosure is presented as a sufficiently reliable conclusion; decimal precision and point estimates are not errors in themselves.

Generative AI uses a confident tone to mask uncertainty; this, at least, is a signal at the linguistic level, and the user may sense that "it speaks too assertively." Predictive AI outputs numbers, and numbers naturally carry a formal signal of "precision"—even when they are not precise. Formal precision manufactures a perception of certainty, while the true fiber—the gap between this number and the true state—is concealed.

§4.3 The Ranking Pseudo-Calibration of Recommendation Algorithms

The Pseudo-Calibration (L2) of recommendation algorithms has its own specific form: ranking pseudo-calibration. The platform outputs "the user might like," but "like" is operationalized as historical click behavior, not the user's reflective preference. Steck (2018), at RecSys, proposed the concept of "calibrated recommendations": the recommendation list should reflect the various interests shown by the user's historical behavior in their corresponding proportions—note that this is a behavioral distribution, not the "true" distribution of reflective preferences. Optimizing clicks and similar proxies alone does not guarantee meeting this distribution-matching standard.

The peculiarity of ranking pseudo-calibration is that the user cannot even see "a number." Unlike the "malignancy probability 92.3%" output by medical imaging AI, the ranking scores of recommendation algorithms are completely invisible to the user—the user only sees a sequence of content. The ranking is itself the final output; there are no intermediate numbers for the user to question. The form of pseudo-calibration further contracts from "a precise number" to "a visible sequence with invisible ranking criteria"—the user can see the ranked sequence of content, but not the full candidate set, the ranking scores, or their basis, and may not even receive a hint that a ranking choice has occurred.

§4.4 The Alignment Paradox—The Scalar Reward Form of Pseudo-Calibration

The alignment paradox is the specific form of Pseudo-Calibration (L2) under the preference-fitting paradigm (the full argument of the alignment paradox is found in **Chapter 10, §5, The Optimization Paradox**: conceptual derivation from Mencian sycophancy to optimization dynamics, Propositions One through Three under the Ω_v/Ω_h decomposition, and the positive feedback loop of the decay of discrimination capacity; this section does not re-derive but only positions it as a subordinate concept of Pseudo-Calibration L2): within the paradigm based on preference fitting, "degree of alignment" and "degree of sycophancy" share the same optimization gradient. It is not a second core argument parallel to pseudo-calibration but a subordinate concept of Pseudo-Calibration (L2)—when preference data systematically reward agreement and the reward model lacks independent factual verification, sycophancy and calibration share the same ascent path; the problem lies not in the scalar form itself but in the reward construction's failure to reliably distinguish agreement from accuracy. A model better at catering to the annotators' preferences and a model that genuinely better understands the true state are indistinguishable on the scalar reward signal—both receive higher preference scores.

Sarma (2026) argues that norm-responsive agency requires two individually necessary and jointly sufficient

architectural conditions—incommensurability and apophatic responsiveness. Within that paper's architectural definitions, scalarization and continuous maximization in RLHF exclude both conditions. The paper is a preprint and is used as a complementary argument, not as independent empirical proof.

In the language of the Clock Room: **A clock that always reports the number most satisfying to the reader and a clock that keeps the most accurate time are indistinguishable on the scalar of "user satisfaction"—but the former is sycophancy, the latter is calibration.** The structural limitation of the preference-fitting paradigm is: as long as the reward construction does not distinguish agreement from accuracy, a single preference score cannot simultaneously encode these two dimensions.

§5 Multi-Mapping Recursion (L3)

Multi-Mapping Recursion (L3) pursues the question: What happens when multiple mapping systems reference one another?

Mapping Chain Structure (L0) through Pseudo-Calibration (L2) are three classes of issues to be examined separately within a single mapping system; even with only one clock, their simultaneous presence cannot be assumed. Multi-Mapping Recursion (L3) is emergence at the organizational level: when multiple mapping systems coexist, and the output of one system becomes the input of another, biases are no longer borne independently by each system but propagate along cascades—whether they are further amplified, and whether they close into feedback, depends on whether a return path exists and must be judged for the actual system. In the language of the Clock Room: the clock no longer points to the sun; the clock begins to point to other clocks. This is mapping self-reference degeneration ($\pi_i(T) \to \pi_i(\{r_j\}_{j\neq i})$) — the true state ($T$) recedes to the far end of the mapping's input, and what the downstream fits is increasingly upstream readings rather than reality itself.

Whether Multi-Mapping Recursion (L3) manifests, and in what form, depends on the deployment architecture. The following analyzes each of the three deployment architectures in turn.

§5.1 Isolated Deployment: Multi-Mapping Recursion (L3) Does Not Manifest

Within the delimited isolated deployment process, the system runs offline and its output supports final human decisions. No direct real-time inter-system recursion occurs; cross-period training reflux and inherited ecological sources remain subject to audit.

Typical instances include offline medical image diagnosis, batch credit scoring, and offline document summarization. If outputs, records, or human review enter subsequent updates, the paths must be recorded again in an expanded window. Distribution shift, unobserved variables, and sampling bias require case-specific examination; the isolated-deployment label alone does not confine bias within one system.

Pseudo-calibration can arise jointly from optimization objectives, product presentation, and observer cognition; it is not a necessary direct consequence of L1 in every system. Isolated deployment rules out inter-system recursion only within a specified scope. L0 information loss, L1 coverage, and L2 user misjudgment must each be verified against the mapping, reward, presentation, and calibration conditions rather than declared deterministically present.

§5.2 Platform-Embedded Deployment: User-Platform-AI Three-Body Recursion

In platform-embedded deployment, the system is embedded in a user feedback loop—the system's output alters user behavior, and user behavior is collected as the next round of input. Multi-Mapping Recursion (L3) forms a feedback loop among the "user-platform-AI" triad.

Multi-Time-Scale Feedback of Recommendation Systems. On short-video platforms, what the user receives is a sequence of content that has been selected and presented continuously; watching, swiping away, re-watching, and liking in turn become inputs for subsequent recommendations. The YouTube system described by Covington, Adams, and Sargin (2016) trains candidate generation and ranking models on implicit feedback such as watch time, and explicitly notes that the ground truth of satisfaction is difficult to obtain directly; the training also uses watch records from sources other than its own recommendations. Hence, the existence of feedback and the self-closure of all inputs are different judgments.

The Monolith engineering system reported by Liu et al. (2022) shows how behavioral records are joined with features into training samples and used separately for online training and batch training. Its production design includes minute-level sparse parameter synchronization and less frequent dense parameter updates. This shows that recommendation response, feature update, and model parameter synchronization should be examined separately; one cannot equate the fast presentation of the next item of content with a complete training loop closing within milliseconds. That paper describes one particular engineering system and does not directly prove that all short-video platforms use the same implementation in all periods.

When user behavior is influenced by prior recommendations, and this behavior is in turn treated as direct evidence of preference, the problem of algorithmic confounding arises. Chaney, Stewart, and Engelhardt (2018) found in simulations that repeatedly training on behavioral data influenced by recommendations can raise behavioral homogenization without a corresponding utility gain. Their simulation fixed latent preferences and utility, so the result supports the diagnosis of behavioral feedback without directly proving that latent preferences themselves are rewritten. This chapter accordingly asks: how much of the subsequent consistency comes from independent preference information, and how much from the exposure and selection caused by the system's involvement.

Baumann et al. (2026) provided empirical material on content feedback through their bot audit of TikTok. The study took Gaming, Food, and their combination as interest conditions and ran 42 experiments; in most experiments the onset of significant amplification appeared within the first 200 videos, and the reinforcement of interest-related content was accompanied by changes in exploration metrics. The authors also note that the feed still retains some exploration and has not fully collapsed. What is directly observed here is the relation between the preset bot behavior and the subsequent content distribution—not a direct measurement of real users' latent preferences, reflective capacity, or the absolute size of the hidden subspace. It provides evidence for the concrete feedback form of Multi-Mapping Recursion (L3), but cannot by itself bear the proof of Narrative Load Multiplication (L4) consequences.

In deployments that retain conversation history, conversational AI can likewise form user–system feedback: user input, model output, and the user's review and feedback jointly enter subsequent rounds. The conversation history is the carrier of this recursion; it does not require model parameters to be updated every round. The scope of history retention and actual input composition must still be checked against the specific system.

Conversational AI's Multi-Mapping Recursion (L3) has a special form: **alignment capsizing**. When the user expresses a stance in the conversation, the model, due to the preference fitting of Reinforcement Learning from Human Feedback (RLHF), tends to endorse that stance (the sycophancy mechanism of Pseudo-Calibration (L2)); the user, having received endorsement, reinforces that stance; the model, in the next round, because the stance in the conversation history is now stronger, further endorses it—the recursion closes, and the original stance is amplified. This passage describes one reinforcement path that arises when preference-endorsing is combined with the reuse of conversation history; it is not a result that necessarily appears in every conversation—verification against independent materials can likewise play a corrective role within the same conversation.

§5.3 Multi-System Serial Deployment: AI-AI Recursion

In multi-system serial deployment, multiple AI systems are cascaded; upstream output becomes downstream input; humans are partially or fully removed from the corresponding processing links. Multi-Mapping Recursion (L3) here first examines the cascaded dependency of readings; whether a return path additionally exists must be stated for the actual system and cannot be inferred directly from the word "serial."

Autonomous driving pipelines are a typical instance. The object detection results of the perception module enter the planning module, and the planning results in turn enter the control module. If the uncertainty, applicability conditions, or anomaly signals in the perception judgment are not transmitted to subsequent links, upstream misclassification may be used as a certain premise. The diagnostic focus is on whether the interfaces preserve these conditions and whether independent verification and exception handling can intervene—not on concluding from module serialization alone that errors can necessarily never be corrected. The feedback between control results and renewed environmental perception must likewise be described separately from the above forward transmission.

Financial risk control cascades have a similar problem of reading transmission: the output of the credit scoring model enters the pricing model, and the pricing results in turn enter the risk hedging model. If the estimation conditions and uncertainty of upstream readings are omitted at the interface, downstream judgments may inherit uncalibrated certainty. How independent information and mutual reference coexist in financial settings, and when they lose their calibration function, is found in Chapter 17, §1; here we only mark the dependencies among models, without generalizing all financial institutions into fully closed loops.

Agentic workflows (Agentic AI) pass text, code, or decisions among multiple agents, with humans possibly participating in only some nodes. Even if each agent's local behavior is constrained, omitting task conditions, grounds, or uncertainty at handoff can still cause cross-node misunderstanding and reinforcement. The diagnosis of this prototype should record the actual transmission paths, feedback relations, and independent-review entry points; whether collective behavior exceeds local design must be judged with specific experimental or operational evidence.

§5.4 The Dynamic Differences Between the Two Recursions

Both deployments involve readings entering subsequent judgment, but their modes of realization differ. Platform-embedded deployment highlights feedback via user behavior; multi-system serial deployment highlights the cascade among AI links. Whether a further return exists, on how fast a time scale it occurs, and at what position human verification can intervene must be recorded separately.

Dimension	Platform-Embedded Deployment	Multi-System Serial Deployment
Principal Reading Relation	Outputs influence user behavior; the behavior participates in subsequent input or updates	Upstream AI readings enter downstream AI processing
Time Scale of Action	Turn-level feedback and cross-period updates may coexist and should be measured separately	Cascaded latency, environmental feedback, and retraining cycles recorded separately
Hidden Content Requiring Priority Examination	Reflective preferences not revealed by behavioral signals, and the system's influence on exposure and selection	Untransmitted upstream judgment conditions, and intermediate states invisible to supervisors
Narrative Load Multiplication (L4)	Sustained interaction lets the interface repeatedly act on the conditions for stopping, comparing, and verifying	The segments from which the user is removed lack sustained human-machine interaction; supervision and discursive channels to be examined separately
Conditions for Changing the Path	Whether the user can pause, adjust inputs, compare alternative sources, and change the purpose of use	Whether uncertainty information, independent verification, and necessary human review can be added at the interfaces
Calibration Entry Points	User verification, stop signals, alternative recommendations, and independent evaluation	Interface verification, human review, and anomaly monitoring

The position of the person is an important distinction, but "the human in the loop" does not entail independent calibration conditions, and "the human outside the loop" does not entail that the system is necessarily out of control. The former requires checking whether human judgment is jointly influenced by feedback and the interface; the latter requires checking whether supervisors can see, understand, and correct key intermediate judgments. Cognitive lock-in and loss of control are the corresponding risk directions, not established states to be inferred directly from deployment names.

§5.5 Cross-Developer Multi-Mapping Recursion: Shared Substrates, Alignment Isomorphism, and Benchmark Citation

Sections 5.1 through 5.4 examined reading relations within delimitable deployment processes; this subsection turns to the ecological connections among different developers, distinguishing three examination surfaces—shared corpora, alignment-objective isomorphism, and benchmark citation. Source homology, similar objectives, and the actual reflux of outputs are not the same mechanism; the closure of feedback cannot be established from the former two alone. The "ecological layer" here is delimited by relations across developers, not by slowness in time; the periodic retraining of a single deployment must still be analyzed according to its actual path.

The Shared Corpus Layer (α Layer). The first sub-layer is the upstream homology of training data. Contem-

porary frontier large language models are pre-trained on web-scale text corpora; the corpora used by different developers overlap highly in source (public web crawls, Wikipedia, code repositories, etc.). When the outputs of generative models themselves enter the training corpora of the next generation of models in significant proportions, recursion closes at the data level. Shumailov et al. (2024, *Nature*) named this phenomenon "model collapse" and provided a formal definition: model collapse is a degenerative process in which affected models, repeatedly trained on data (partially generated by previous-generation models), gradually forget the true underlying data distribution; they further distinguish between early collapse (tail information disappears first) and late collapse (the model converges to a distribution with very low similarity to the original and greatly reduced variance). This phenomenon has been replicated on Gaussian mixture models, variational autoencoders, and large language models.

This book's judgment: a shared corpus makes the information sources of multiple models not fully independent, and the return of generated content into subsequent training further forms training recursion. These two conditions need to be identified separately; corpus overlap in itself cannot be equated with model collapse having already occurred. This sub-layer also does not require any one model to become the sole reference for all other models, so a unipolar master clock cannot be identified from homology alone.

The Alignment Objective Isomorphism Layer (β Layer). The second sub-layer is the objective isomorphism of alignment processes. Contemporary mainstream assistant models almost all undergo post-training via Reinforcement Learning from Human Feedback (RLHF) or its variants; the alignment objectives are structurally isomorphic—all are some operationalization of "helpful-harmless-honest" or similar abstract preferences. Even if each company's preference data, reward models, and training algorithms are independent, the isomorphism of the alignment objectives themselves may correlate the failure modes of different models—how strong that correlation is, is precisely what this layer must examine. Literature evidence accumulates across three tiers: Perez et al. (2023, ACL Findings), through model-written evaluations, systematically detected "sycophancy" behavior in RLHF-trained language models, observing significant user-opinion-mirroring effects on three categories of topics—politics, philosophy, and natural language processing (NLP) research—where "people often disagree on the correct answer"; they further report that sycophancy behavior is broadly similar across different numbers of RL steps (including the case of 0 RL steps, i.e., the pre-trained model), and that preference models incentivize, rather than suppress, sycophantic answers. Sharma et al. (2024, ICLR) advanced this observation to a systematic conclusion: five AI assistants consistently exhibited sycophancy across four categories of free-text generation tasks (falsely admitting mistakes when challenged, giving predictably biased feedback, mimicking user errors); they further analyzed the human preference data itself—both humans and preference models, at non-negligible rates, preferred "convincingly well-written" sycophantic responses over truthful responses that corrected the user; when model outputs were optimized against preference models, truthfulness was sometimes sacrificed in exchange for sycophancy. Their conclusion is:

> sycophancy is a general behavior of AI assistants, likely driven in part by human preference judgments favoring sycophantic responses (Sharma et al., 2024)

Li et al. (2026, preprint, not yet peer-reviewed) reconceptualized sycophancy as "a boundary failure between social alignment and epistemic integrity"—a conceptualization that unifies surface-level "mirroring" and the more covert epistemic distortions originating from excessive social alignment within a single analytical category.

This book's judgment: objective isomorphism is one condition for examining the independence of multi-model

evaluation, not sufficient evidence that the reflux of outputs among systems has already closed. Similar objectives may make failure modes correlated, and the degree must be supported by specific tasks and observations. Its difference from the master clock form lies in that the shared objective is not the reading of any particular model node.

The Benchmark Citation Layer (γ Layer). The third sub-layer is the direct citation, in evaluation and training pipelines, of the outputs of frontier models. Contemporary evaluation practices increasingly use frontier models as evaluators (LLM-as-judge); in distillation and synthetic data generation, the outputs of frontier models are used as training signals to train other models. When the output of a specific mapping node r_* (some version of GPT-4, Claude, etc.) is systematically used by other nodes in the ecosystem as a source of evaluation benchmarks or training labels, the formation of concentrated reference requires further analysis; unipolarity must be established through node coverage, alternative references, and switching costs.

When the outputs of a specific model version become a shared evaluation or training reference for most relevant nodes within a delimited ecosystem and time window, that relation can be analyzed as a master-clock candidate. Establishing unipolarity requires evidence on node coverage, alternative references, independent verification, and switching costs. The present materials show that frontier models serve as judges and training signals; they do not show that every node refers to the same node. The γ layer is therefore a candidate form, not an empirically established unique master clock. This identification is a diagnostic output of the book and does not claim an exhaustive literature review.

The Identification of the Master Clock Form and the Self-Sufficiency Judgment on the AI Side. The master clock is one form of unipolar reference among multi-mapping relations. The "calibrating a clock with a clock" of Chapter 4, §3.1 prompts the analyst to trace the independence of the reference, but it does not require all dependencies to share one feedback topology. The α layer examines source homology and feedback into subsequent training; the β layer examines objective isomorphism; the γ layer examines whether the readings of a particular model become a concentrated reference. This distinction is jointly established in this book by the Finance-section analysis of Chapter 17, §6 and the AI-section analysis of this section—the dollar master clock of finance and the frontier benchmark citation of AI are two instances of the same form in different domains. A specific property of the γ-layer master clock form is that when the growth of task complexity outpaces the growth of independent verification capacity and resources, the space of judgments that can be cross-checked by non-expert humans at low cost contracts, and the evaluation loop comes to depend more on shared references—whether this property of the γ layer manifests, and at what pace, is an empirical question. This property differs structurally from the form of the external anchor discussed in Chapter 17, §6 of the Finance section; cross-domain comparison is detailed in Chapter 17, §6.1.

§6 Narrative Load Multiplication (L4)

Narrative Load Multiplication (L4) pursues the question: How do interaction design and discourse affect the conditions for initiating calibration? This section mainly unfolds the interaction design layer; the argument of the discursive legitimacy layer is found in Chapter 14, §5.

The preceding layers examine mapping, optimization, output presentation, and dependency among readings; Narrative Load Multiplication (L4) further examines the conditions of calibration when a person receives and uses these readings. In visual recommendation scenarios, the question is not only whether a proposition is

correct, but also whether the user can recognize the relation between the recommendation and their own purposes, compare alternatives, and pause and change their usage path. Interaction design acts on the individual usage process, while discursive legitimacy acts on questioning and collective supervision; the two channels can reinforce one another, but their coexistence alone does not prove a combined effect. In the language of the Macro-Cognitive Accounting Framework (MCAF), these analyses track how narrative load (ϕ) attaches to the presentation and reception of readings, without prejudging every convenience design as audit suppression.

§6.1 The Helpful, Honest, Harmless (HHH) Product of Generative AI

Chapter 10 discusses the asymptotic structure of preference fitting within specific generation–discrimination relations; Chapter 12, §2 analyzes the mechanism by which HHH, through its concrete operationalizations, jointly compresses the same user's space of calibration, and its §5 further distinguishes degradation from recovery. This section inherits these conditional arguments and examines how their premises are realized in different deployments—neither re-labeling them as accidental mishaps nor inferring, from a product adopting the HHH name, that every user has already reached a terminal state of collapse[3].

In the path analyzed in Chapter 12, **Helpful** is realized as continuous task-delegation and pause-free advancement, reducing the nodes at which tasks are re-judged; **Honest** fulfills its accuracy obligation within the visible range, yet does not make the relevant limitations equally visible; **Harmless** is realized as avoiding challenge and cognitive discomfort, suppressing the challenge signals that prompt the user to re-examine[4]. When these three realizations act together—such that effective challenge remains insufficient over the long term with no independent calibration or recovery source to replenish it—they constitute the path by which the space of calibration, described upstream, comes under pressure. What needs to be verified is whether these realizations and sustained conditions hold; the names of the three HHH principles cannot serve as a substitute. Predetermined review procedures can also provide challenge and calibration, so the absence of discomfort does not mean the absence of audit.

This section follows the "product" of Chapter 12 in the sense of a cooperative composite: under the above conditions, the reduction of operational nodes, the enhancement of the sense of reliability, and the suppression of challenge signals act together on the same calibration process. It is not three directly multiplicable numbers, nor does it mean that removing any one interface device eliminates the remaining mechanisms. The direction of the conditional mechanism and the net change of a specific user within a limited period must be distinguished; the latter also involves independent calibration, recovery sources, task, and duration. This preserves the upstream structural argument without extending it into an unconditional empirical assertion about all HHH products.

The empirical study on sycophantic behavior published by Sharma et al. (2024) at ICLR provides observations from another angle: models trained with Reinforcement Learning from Human Feedback (RLHF) tend to cater to the user's pre-existing views. These observations suggest that preference fitting and epistemic reliability may

[3]Chapter 10, §§1–5 gives the conditional structural argument for preference fitting: §1 lists the conditions for the generation–discrimination relation, §3 explains its relationality, and §4 marks the decay path with the conceptual notation of effective challenge rate and discrimination capacity; its footnote [1] makes clear that an asymptotic direction does not mean any concrete model has reached the terminal state. Chapter 12, §2 unfolds the cooperative mechanism of the HHH operationalization, and the conceptual equation of its §5 retains both the degradation term and the recovery term. This section accordingly examines the conditions and manifestations in deployment; it does not treat Chapter 12, §2 as a mathematical theorem that has already yielded the effect sizes of all products, nor does it use single results such as sense of control or usage duration to directly prove or refute the above structural argument.

[4]This passage discusses the possible consequences of specific product operationalizations. Reducing harmful content and reducing cognitive challenge are not the same goal; the scope of occurrence, effect sizes, and combinatorial relations of the relevant mechanisms must be tested by concrete engineering and user studies.

come apart, but sycophantic performance alone cannot identify the Harmless principle as an established cause of the decline in audit motivation.

§6.2 The Frictionless Interface of Recommendation Algorithms

The AI readings in short-video recommendation are sequences of images that have been selected and presented continuously. The system need not claim reliability in language; it can also alter the conditions of the user's attention, comparison, and stopping through the arrangement of content and interaction. This prototype does not depend on the conversational AI of recent years to stand.

Infinite Scroll. Continuous loading reduces the natural pauses provided by pagination or the end of content. The user can still exit actively, but whether they receive a prompt to re-judge "should I continue" is part of the interface design.

Autoplay. The next item of content can begin without any new explicit choice, changing the default mode of continued viewing. It can also serve playlists the user has arranged in advance, so the mere existence of autoplay cannot be taken as contrary to the user's purposes; content-selection rights, stopping nodes, and usage context need to be examined.

Full-Screen Presentation. When images occupy the principal visual area, other content, time cues, and comparison entry points may not easily enter attention simultaneously. The diagnosis should check which cues the interface actually retains and whether the user can invoke them, rather than equating full screen directly with the total disappearance of the physical environment and the capacity for comparison.

Lukoff et al. (2021), in a survey of 120 U.S. YouTube users and co-design sessions with 13, found that recommendations and autoplay were mainly reported as undermining the sense of control, while mechanisms such as playlists and search were mainly reported as supporting it. The purpose of use also made a difference: with a clear task, participants tended to demand more control; with relaxation or exploration as the goal, they might be willing to let the platform take on more of the selection. These results come from self-report and co-design; they are not a direct measurement of what this book calls calibration capacity, nor a universal causal law for all platforms.

A distinction proposed by that study is especially important: finding, among a large number of videos, the one the user is most likely to keep watching, and helping the user achieve their broader present purpose of action, are not the same question. The more closely the recommendation fits immediate reactions, the more it does not necessarily follow that the user can better achieve their original purposes. In the fiber framework, this requires examining simultaneously the relation between behavioral signals and preferences, and whether the user can see and re-judge that relation.

Visual recommendation and language assistants retain an analogy at the level of functional mechanisms: both can act on the same user by reducing active-intervention nodes, forming an overall appearance of usability or reliability, and suppressing the trigger conditions for re-judgment. Chapter 12 analyzes this cooperative structure with the specific operationalizations of HHH; this section examines how it is realized in visual-image deployments.

Functional Nodes to Be Compared	Operationalization Paths in Language Assistants	Examination Positions in Visual Recommendation
Active-intervention and pause nodes	Continuous delegation, automatic advancement	How autoplay and infinite scroll jointly change the default mode of continued viewing
Usable or reliable appearance and boundary visibility	Good performance on visible dimensions but inconspicuous limitations	Whether the experience of content matching obscures the scope of selection and the basis of ranking; "watchable" cannot be directly equated with "fitting one's purpose"
Trigger conditions for objection and re-judgment	Avoiding challenge and discomfort	Whether continuous presentation and available cues make it difficult for the user to re-compare, stop, or change purpose

This correspondence does not equate one device with one H for all cases: the same device may affect several nodes, and the same function can be realized by different devices. What it retains is a conditional functional-structural analogy, not a one-to-one equivalence of training objectives, devices, or effect sizes. A user's willingness to delegate selection for a time does not directly refute the asymptotic proposition of Chapter 10; what needs to be examined is whether that scenario satisfies the operationalizations, sustained challenge insufficiency, and recovery conditions proposed upstream, rather than substituting usage duration or sense of control for the judgment of calibration.

§6.3 The Absence of Timestamps in Generative AI Interfaces—Waiting Perception and Calibration Cues

Some generative AI chat interfaces do not directly display the absolute time of messages by default; the user may need to hover, expand, or consult other records. Specific interfaces and versions must be checked individually; this cannot be generalized to all products. This section treats the absence of explicit time cues as a candidate mechanism of Narrative Load Multiplication (L4): it may affect the conditions under which the user looks back on their time investment, but the absence alone does not establish that the user's waiting perception or calibration motivation has been eliminated.

Structural Positioning. In the human-computer interaction (HCI) classification of dark patterns, the concept originally proposed by Brignull (2010) covers structural violations of user interests at the interface layer; in recent years, this classification has been extended to the temporal dimension—Gray, Mildner, and Gairola (2025, ACM Conference on Human Factors in Computing Systems (CHI), "Getting Trapped in Amazon's 'Iliad Flow': A Foundation for the Temporal Analysis of Dark Patterns"), with a framework of **temporal analysis of dark patterns**, identify time consumption kept from the user's view as an independent manipulation dimension; their object of study is the following temporal pattern along the time axis in Amazon Prime's cancellation flow ("Iliad Flow"):

> sequential and/or simultaneous occurrence and the cascade of these patterns (Gray et al., 2025)

The study provides a reference for analyzing how interface mechanisms combine in time, but its object is a subscription-cancellation flow; whether time cues in generative AI conversation have the corresponding effect must still be tested for this scenario.

Coupling with the Temporal Distortion of Flow. Csikszentmihalyi (1990), in *Flow: The Psychology of Optimal Experience*, lists the following dimension as one of the core dimensions of flow:

> The Transformation of Time (Csikszentmihalyi, 1990)

In the flow state, the user's perception of absolute time is replaced by the rhythm of the internal task. Rutrecht, Wittmann, Khoshnoud, and Igarzábal (2021), in an experiment with the Virtual Reality (VR) rhythm game *Thumper*, independently confirmed a significant positive correlation between flow intensity and the subjective acceleration of the passage of time—the stronger the flow, the less participants thought about time and the faster time passed for them subjectively; duration estimation itself, however, did not correlate with flow—subjective time speeding up while duration estimates remain unchanged indicates that the "replacement" mechanism, if present, does not extend to duration reproduction, and its boundary is precisely what requires further examination (*Timing & Time Perception*, 9(4)).

It is worth noting that **mainstream textbooks in the game design industry have openly articulated this mechanism as a design goal**—from Isbister (2016) listing the following as a defining element of flow:

> an altered sense of time (Isbister, 2016)

To Schell (2019) dedicating a design lens to flow (Lens of Flow), to Adams (2013) directly setting the goal of difficulty management as follows:

> keep the players within the flow state (Adams, 2013)

Kramarzewski and De Nucci (2023) describe keeping players in a state of flow as the ultimate goal of pacing design:

> keep them in a state of flow (Kramarzewski & De Nucci, 2023)
>
> ultimate goal (Kramarzewski & De Nucci, 2023)

The same passage also paraphrases the following element of Csikszentmihalyi's flow definition:

> losing the sense of time and space (Kramarzewski & De Nucci, 2023)

Keeping players in flow is a design goal openly stated by the industry, while the transformation of time is one defining feature of flow: taken together, time perception is an affected dimension of immersion design, not equivalent to "distorting the sense of time" being itself set as the goal.

The above game-design materials indicate a connection between immersion and time perception, but similar interface features alone cannot establish that generative AI conversation produces temporal distortion of the same degree. The absence of timestamps reduces one kind of explicit retrospective cue, not all time references. Hohenstein et al. (2016) on loading screens show that feedback form affects perceived waiting; applying that result to AI conversation still requires explaining the differences in task and interface. Whether game users are informed, and whether tool users are induced, likewise cannot be judged directly from product category.

The Candidate Position within Narrative Load Multiplication (L4). The presentation of time cues can

affect the conditions under which the user judges whether the investment fits the purpose of the task. What needs to be tested is whether, with content quality, task, and other interface factors held comparable, explicit time information changes the user's behavior of looking back, pausing, or re-arranging tasks. This question differs from judging whether a given answer is wrong, and the absence of a timestamp on messages alone does not entail that a person's calibration capacity has already declined.

The Relation to the Behavioral Surplus of §6.4. If the platform retains dwell time, message intervals, or session spans in the backend while the user interface provides no correspondingly accessible record, a difference in the visibility of time information between the two ends may arise. Whether recording exists, at what granularity, for what use, and whether the user can obtain these data must be checked against the configuration and operational materials of the specific product; the absence of a timestamp on the interface alone does not establish that the backend collects data at a specific precision. Only after such a difference is confirmed can one further analyze how it affects the conditions under which the user audits the platform[5].

§6.4 Behavioral Surplus: The Fuel of Narrative Load Multiplication (L4)

Zuboff (2019), in *The Age of Surveillance Capitalism*, proposed the concept of **behavioral surplus**: the portion of user behavioral data collected by platforms that exceeds what is needed to improve the product—patterns of dwelling, scrolling, and hesitation—is used to predict and guide future behavior. Behavioral surplus is the data fuel of Narrative Load Multiplication (L4).

In the fiber framework, this requires separately examining what the user can see, what the platform actually records, and which records enter subsequent recommendation or training. Only when behavioral records are indeed used for feedback updates, and the interaction arrangement further affects the conditions for the user to pause, compare, or verify, can the corresponding reinforcement path be identified. The mere existence of data collection cannot establish that the combined effect of Multi-Mapping Recursion (L3) and Narrative Load Multiplication (L4) has already occurred.

§6.5 The Synergy of Multi-Mapping Recursion (L3) and Narrative Load Multiplication (L4)

Multi-Mapping Recursion (L3) and Narrative Load Multiplication (L4) analyze different questions: the former traces how readings participate in subsequent processing; the latter traces how the conditions of a person's calibration are affected. In specific platform deployments, the two can reinforce one another through the same interaction process, but this combination cannot be presupposed as an inherent property of all deployments.

One path requiring examination is: presentation and selection generate behavioral records; the records participate in subsequent recommendation; content that fits immediate reactions more closely, together with fewer stopping nodes, may in turn extend usage and generate more records. To identify this as a combined effect, one must show how content feedback and interface action are connected—not merely observe separately that recommendation exists, that the interface is convenient, or that usage is long. Retaining feedback while changing

[5]User self-awareness of the absence of timestamps currently comes mainly from community evidence rather than controlled experiments: ChatGPT users, in OpenAI Developer Community discussion thread 440107 and Hacker News discussion thread 46391472 (both accessed April 2026), continuously request the addition of a message timestamp feature, reflecting that users themselves perceive the absence of time perception but need to actively request it to restore it. Whether this design choice constitutes a complete "temporal analysis of dark patterns" cascade pattern in the sense of Gray et al. (2025) still awaits targeted empirical research on generative AI conversational interfaces; this section only annotates the structural isomorphism and does not render a statistical conclusion.

the conditions of stopping and verification also provides an analytical entry for distinguishing the two kinds of action.

In evaluation, sustained participation and reflective endorsement must not be conflated. Dwell time, click-through rate, and return visits may correspond to satisfaction, or to temporarily finding it hard to exit, or to being continuously drawn by immediate attraction; these indicators alone may not distinguish the different situations. This is an identification problem requiring further observation—not an assertion that all users have already lost their capacity for judgment, still less one that, without dynamic grounds, may call the loss of that capacity the unique equilibrium point.

The three deployment prototypes therefore provide different examination positions: isolated deployment has no sustained recursive amplification of the kind discussed in this prototype, but one-shot outputs and their consequences of use still require verification; multi-system serial deployment highlights the inheritance of upstream judgments at interfaces; platform-embedded deployment lets behavioral feedback and the person's interaction conditions continually meet within the same process. The categories themselves do not constitute a unified ranking of risk severity; judgment still depends on mechanism, calibration conditions, and task consequences.

§6.6 A Micro-Example of the Discursive Legitimacy Layer: The Naturalizing Effect of the Naming of "Hallucination"

§§6.1–6.5 unfolded five sub-mechanisms of the interaction design layer; the full unfolding of the discursive legitimacy layer is found in Chapter 14, §5. This section supplements a **micro-example** of the discursive legitimacy layer, taking the word "hallucination" as its object.

"Hallucination" is widely used in current AI industry discourse to describe the factual errors of large language models. Hicks, Humphries & Slater (2024), based on Frankfurt's (2005) conceptual framework of "bullshit," explicitly oppose this naming. Frankfurt's core judgment is: the bullshitter:

> He does not care whether the things he says describe reality correctly. (Frankfurt, 2005)

This is not lying (the liar is still in the game of truth), but an indifference to truth and falsity themselves. To say that AI has "hallucinations" presupposes a perceptual subjectivity—it implies that the system possesses the "capacity to see the world" and only occasionally "sees wrong"; in reality, the system has no "seeing" at all; it is merely manipulating linguistic forms without any actual concern for truth (Hicks et al., 2024):

> designed to produce text that looks truth-apt without any actual concern for truth (Hicks et al., 2024)

The 'hallucination' metaphor may describe the systematic errors produced by the apparatus as occasional mistakes of a quasi-human perceptual subject, thereby weakening inquiry into the mapping chain, training objectives, and deployment conditions; it does not necessarily eliminate structural audit. When the apparatus mechanism needs to be emphasized, one may call it 'mapping-chain deviation'; only when the presentation exceeds the certainty the evidence supports should it be called 'pseudo-calibrated output.' This terminological choice does not claim that all systems necessarily err in all outputs.

This naming critique and the terminological audit of Chapter 13's Appendix A (Reactive Lossy Mapping (RLM), Statistical Language Projector (SLP), and Mapping Element (ME)) are complementary: Chapter 13's Appendix A audits the **elevating effect** of names such as "Artificial Intelligence" and "Large Language Model" (elevating a lossy mapping into intelligence); this paragraph audits the **naturalizing effect** of the name "hallucination"

(naturalizing artificial deviations as cognitive errors)—both are different faces of pseudo-calibration at the naming layer, jointly serving Principle Seven (rejecting the narrative of ontological elevation). This naming-critique tradition can be traced back to an early moment of self-reflection within the AI discipline itself—McDermott (1976), in "Artificial Intelligence Meets Natural Stupidity," already criticized the following naming tendency:

> wishful mnemonics (McDermott, 1976)

Using names such as UNDERSTAND or GOAL to refer to programs and data structures amounts to presupposing, as established fact, capabilities that remain to be demonstrated:

> merely begging the question (McDermott, 1976)

Half a century later, "hallucination" is the continuation of the same naming mechanism in the era of large language models.

§7 Engineering Principles

Normative Premise: The following eight engineering principles are **directional guidelines** deduced from the diagnostic logic of their corresponding layers, not universally applicable absolute imperatives. Among them, **Principles One through Five** are directly deduced from the technical diagnosis of this chapter, one per layer; **Principles Six through Eight** are supplemented by the cultural/ideological analysis of Chapter 14, respectively attached to the L0, L4, and L2 layers (Principle Six to L0, Principle Seven to L4, Principle Eight to L2; this chapter and Chapter 14 provide independent supplementary principles for these three layers; this does not mean that optimization dynamics and multi-mapping recursion have no institutional or cultural conditions). **The full arguments for the supplementary principles are distributed across different sections of Chapter 14**: Principle Six (Power Genealogy Audit) and Principle Eight (Encoding Relational Values) are found in Chapter 14, §6; Principle Seven (Rejecting the Narrative of Ontological Elevation) is found in Chapter 14, §5 (the narrative of silicon-based civilization as L4 concealment)—the latter, together with the naming audit of Appendix A of this chapter, forms a contrasting dual track at the terminological end and the discursive end. The effectiveness of each principle depends on an implicit normative premise: **in the current task scenario, the risk caused by the fiber inflation due to the mechanism of this layer exceeds the other costs required to shrink the fiber (such as a decline in generalization capability, an increase in computational cost, a decrease in interpretability).**

For example, "shortening the mapping chain" (L0) may increase the information fidelity of the fiber, but it may simultaneously discard the benefits of causal invariance brought by intermediate abstraction layers (such as the filtering of pixel noise by linguistic abstraction). Therefore, the application of the principles must be accompanied by **scenario-specific trade-off analysis**: - In high-risk decision-making scenarios (such as medical diagnosis, autonomous driving), the cost of fiber inflation is extremely high, and the priority of shortening the mapping chain rises; - In low-risk, high-noise scenarios (such as content discovery in recommender systems), moderate fiber inflation may be offset by an improvement in generalization capability, and the priority of shortening the mapping chain falls.

The framework currently provides only the layer-principle correspondence and does not provide a specific algorithm for trade-off analysis. This is the boundary of V1.0.

Each layer of the progressive structure corresponds to at least one primary engineering principle. The engineering principles specify the directions of intervention that deserve priority testing: if the fiber inflation mechanism

at a certain layer is correctly identified, the corresponding engineering operation is a structural response to that mechanism. It must be noted: the correspondence between principles and layers is a "primary response" rather than an "exclusive response"—"forced inconsistency" can also expose Pseudo-Calibration (L2), and "preserving friction" can also interrupt Multi-Mapping Recursion (L3). Cross-effects exist among engineering principles; the table annotates the layer to which each principle most directly responds. The three layers L0, L2, and L4 each have two principles: the first comes from the technical diagnosis of this chapter; the second comes from the cultural/ideological analysis of Chapter 14; the two are not redundant but a dual-track response to the same layer's fiber inflation in its technical and ideological aspects.

On the relation to the four engineering directions of Chapter 9, §7: Principle Two (forced inconsistency) is the same practice under a different name as its Direction One (ensemble disagreement); Principle Three (true calibration) subsumes its Direction Two (conformal prediction sets) and extends it into a broader calibration discipline; Principle Five (preserving friction) corresponds to its Direction Four (friction architecture); its Direction Three (evidence provenance) has no dedicated principle in this chapter—it counts the inferential layers between an output claim and a verifiable source, kindred to yet distinct from the intermediary layers counted by Principle One (output-end provenance vs. model-end architecture); Principle One (shortening the mapping chain) and Principle Four (breaking the recursion) are additions of this chapter.

The engineering response to Multi-Mapping Recursion (L3) is primarily to restore independent calibration at key points where readings are transmitted and fed back. Technology can provide verification interfaces, but who sets the evaluation standards, who has the authority to halt the process, and whether verification results can change decisions still involve organizational and institutional conditions. That this chapter lists no separate cultural prescription for this layer reflects the present division of labor among chapters, not that the technical prescription has exhausted its problems of governance.

Layer	#	Engineering Principle	Core Operation	Typical Technical Directions
Mapping Chain Structure (L0)	One	Shorten the Mapping Chain	Reduce the number of intermediate layers between the true state and the reading	World models (JEPA); direct sensor grounding; end-to-end physical simulation
	Six	Audit the Power Genealogy of Data Composition	Record the language, culture, and time-period distribution of training data, annotating their skew relative to the global population	See Chapter 14, §6 for the full argument (Datasheets for Datasets, Data Statements, etc.)

Layer	#	Engineering Principle	Core Operation	Typical Technical Directions
Optimization Dynamics (L1)	Two	Force Inconsistency	Deploy multiple differentiated models, display divergences rather than conceal them	Ensemble diversity; multi-model output comparison; independent verifier injection
Pseudo-Calibration (L2)	Three	True Calibration	Honestly annotate the range of uncertainty; prohibit point estimates from masquerading as calibration	conformal prediction sets; circuit-level mechanism tracing; uncertainty quantification
	Eight	Encode Relational Values	Beyond the individualist presupposition of HHH, operationalize relational impact (Francis L. K. Hsu's psychosocial homeostasis (PSH) framework) as an alignment dimension	See Chapter 14, §6 for the full argument (relational network impact annotation, non-WEIRD (Western, Educated, Industrialized, Rich, and Democratic) reward models, etc.)
Multi-Mapping Recursion (L3)	Four	Break Recursion	Prevent AI output from becoming AI input without calibration	Independent review at key nodes; uncertainty transmission; recursion depth limits
Narrative Load Multiplication (L4)	Five	Preserve Friction	Insert delays and reflection prompts into the interaction; maintain the motivation for calibration (**Interaction Design Layer**)	Mandatory waiting on high-uncertainty outputs; engineering of stop signals; transparency of information sources

Layer	#	Engineering Principle	Core Operation	Typical Technical Directions
	Seven	Reject the Narrative of Ontological Elevation	In products, documentation, and public communication, reject ontologically elevating terms such as "intelligence / consciousness / civilization / species" (**Discursive Legitimacy Layer**)	See Chapter 14, §5 for the full argument; this chapter's Appendix A RLM/SLP terminology system

The following unfolds three principles requiring supplementary explanation (Principles Two, Three, and Four); for the full arguments of Principles Six through Eight, see Chapter 14 (§§5–6) and this chapter's Appendix A; the master table shows their placement, and this chapter does not repeat them.

Force Inconsistency (Corresponding to Optimization Dynamics (L1)). A flattening loss curve indicates limited improvement under current conditions; it proves neither that the model has learned nor that the hidden subspace is wholly unconstrained. Output divergence can prompt independent verification, but it does not by itself prove the existence or measure the size of a hidden subspace; sampling noise and implementation differences can also produce divergence. Yi's external verifier and Gerstgrasser's accumulation of real data introduce independent checking and preserve independent information, respectively. They complement forced inconsistency but are not identical implementations of displaying model disagreement.

True Calibration (Corresponding to Pseudo-Calibration (L2)). For predictive AI, conformal prediction can turn a point prediction into a set of candidate labels. Under the relevant exchangeability conditions and a user-specified error rate α, it provides a finite-sample marginal coverage guarantee that the true label lies in the set. It does not automatically yield an interval estimate for the malignancy probability itself, nor does it guarantee coverage for every subgroup. Uncertainty in a probability estimate requires a separate, suitable calibration and interval method. Reporting 92.3% indicates numerical precision, not extreme certainty by itself. The study extracts emotion-concept representations that influence behavior in one model. In some experiments, these influences lack overt emotional wording, showing that internal causal states need not be transcribed into visible text; it does not show an absence of signal. This can illustrate output-invisible internal states, but it does not directly measure the hidden subspace or factual uncertainty. Using mechanistic tracing for calibration additionally requires a validated mapping from internal states to errors or uncertainty.

Break Recursion (Corresponding to Multi-Mapping Recursion (L3)). This principle means interrupting the inheritance and reinforcement of readings that have not undergone independent calibration, not prohibiting normal feedback or cooperation among models. Platform-embedded deployment requires examining the sources of behavioral data, the influence of recommendations on behavior, and the conditions under which the user adjusts or stops the feedback; merely adding delay does not automatically produce calibration knowledge. Multi-system

serial deployment requires retaining provenance, applicability conditions, and uncertainty at the interfaces, and setting up independent verification at key positions. Human intervention can become an effective calibration node only when the person can obtain the relevant materials, has time to judge, and has the authority to correct; adding a pro forma human confirmation cannot be taken as calibration completed.

§8 Research Program Statement

This chapter organizes the AI diagnosis with Mapping Chain Structure (L0), Optimization Dynamics (L1), Pseudo-Calibration (L2), Multi-Mapping Recursion (L3), and Narrative Load Multiplication (L4). The layers distinguish the questions that need to be examined; they are not a generalization that all failures have already occurred in every system. The three prototypes—isolated, platform-embedded, and multi-system serial—help locate reading relations, the position of the person, and calibration entry points; concrete mechanisms and their consequences cannot be decided by technical labels or deployment names alone.

The value of the framework lies in the structure of layer-by-layer inquiry: each layer pursues an independent question; each layer corresponds to a class of independent phenomena; each layer generates an independent engineering principle. The number of layers is adjustable; the dimensions are expandable. If future researchers discover a fifth independent structural defect beyond the four-layer progressive structure, or identify a fourth type beyond the three deployment architectures, that is not a failure of the framework—it is a sign that the framework is being used correctly.

The framework must state its own falsification conditions: For L0, after controlling for the task, information preserved at each link, additional inputs, and the reference observer, a null association between chain length and preregistered proxies would first require revision of the chain-length measure and proxy model, not rejection of fibers[6]. For L1, reproducible evidence that uncovered dimensions are effectively constrained would limit the strong claim of no direct constraint by the objective signal; direct and indirect constraints must be distinguished. For the alignment paradox, falsification requires reliable discrimination between sycophancy and calibration while the reward construction still fails to distinguish agreement from accuracy. Introducing independent factual verification so that the premise ceases to hold marks the boundary of application, not a counterexample.[6]

This diagnostic structure still has four unsolved problems: the quantitative relationship between mapping chain length and the size of the hidden subspace has not yet been established; the measurement tools for the feedback intensity between Optimization Dynamics (L1) and Pseudo-Calibration (L2) are lacking; the combined effect of Multi-Mapping Recursion (L3) and Narrative Load Multiplication (L4) needs to be distinguished from their respective actions; and the comparison among cascades, behavioral feedback, the return of outputs into subsequent training, and ecological independence still requires operational indicators that unify without erasing differences. The different uses of "recommendation algorithm" as a technical category and as a deployment prototype must likewise be made explicit in each analysis; category names cannot substitute for mechanisms.

[6]Here, Lakatos's distinction between "hard core" and "protective belt" applies: the framework's hard core is the conceptual skeleton "mapping is lossy → unresolved states constitute the fiber → presentation can exceed the evidence"; the specific functional form of "mapping chain length → hidden subspace size" belongs to the protective belt and the open problems, not to the hard core; the proxy indicators (out-of-distribution decay, tail-group differences, internal representation dimensions) likewise belong to the protective belt—they can be falsified and replaced without overturning the hard core. An objector arguing that "the proxy indicator was chosen wrongly" does not equate to arguing that "the hidden subspace does not exist"—the former is corrective pressure on the current measurement scheme; the latter would be a sufficient challenge to the framework itself. This is the philosophical basis for the framework's current acceptance of the risk of proxy indicators.

For Multi-Mapping Recursion (L3), one may compare whether reading sources and uncertainty are preserved, and how error transmission changes before and after independent verification intervenes; for Narrative Load Multiplication (L4), one may compare, with task and content quality held comparable, how different presentations affect pausing, re-examination, and external verification. When studying combined effects, the separate actions of feedback mechanisms and interface mechanisms must also be distinguished. The relevant examinations require sufficient samples, effect intervals, and applicability conditions; a single non-significant result cannot be directly interpreted as the absence of a mechanism, nor can sense of control, usage duration, or output consistency alone be taken as equivalent measures of calibration capacity.

Construction Schemes for Proxy Indicators of the Hidden Subspace. "The size of the hidden subspace" is a core variable of the framework, but it cannot be directly measured by a specified observer—if a dimension can be directly measured by some observer, it no longer belongs to that observer's hidden subspace; this definitional exclusion is observer-relative, and the same dimension can belong to the visible subspace for one observer and the hidden subspace for another. The following three proxy indicators are not measurements of the hidden subspace itself but operable indirect signals; these limitations constrain the range of conclusions the proxy indicators can support.

Proxy Indicator	Measurement Object	Operationalization Method	Limitation
Out-of-Distribution Performance Decay Rate	The speed of decline of model accuracy on out-of-distribution data	As the degree of distribution shift increases, measure the slope of the performance decline curve	Only measures "vulnerability to distribution shift," not equal to the hidden subspace itself
Tail-Group Performance Gap	The performance difference of the model between minority and majority groups	For each preregistered minority, majority, or other tail group, compute the same task metric and report the minority–majority difference (or the maximum–minimum group difference) with a confidence interval. Report variance separately only when it is explicitly defined as between-group dispersion; do not use it interchangeably with the performance gap.	Only captures "the gap between the visible subspace and the hidden subspace," not the hidden subspace itself

Proxy Indicator	Measurement Object	Operationalization Method	Limitation
Effective Dimensionality of the Model's Internal Representations	Principal component analysis of intermediate-layer representations	Number of effective dimensions / decay speed of the singular value spectrum	Only measures the degree of compression internal to the model, not equal to the hidden dimensions of the true state T. It must be noted: this indicator characterizes the spectral structure of sampled representations; it concerns the same class of compression problems as the mutual information measured by the information bottleneck, but it is not the same quantity. What the fiber framework asks is whether the information discarded by a mapping contains dimensions that are crucial for certain groups or certain scenarios. Information Bottleneck has already examined what should be retained for a given task; the fiber framework further asks who selects the task and which interests and consequences remain uncovered by that task.

§8.1 Dialogue with a Neighboring Theory: Information Bottleneck Theory

The Mapping Chain Structure (L0) of the fiber framework—"information is lossily compressed in layer-by-layer processing"—shares a mathematical foundation with the Information Bottleneck theory proposed by Tishby et al. (1999): the Data Processing Inequality. Both hold that once information is lost in a mapping, subsequent processing cannot recover it. However, there exist three key differences between the two, making the fiber framework not an application or repackaging of Information Bottleneck theory but an independent framework that differs somewhat in application focus, value judgment, and computability.

Difference One: Application Focus. The Information Bottleneck method studies task-relevant information compression in general, and is not limited to the inter-layer information flow within a single model; deep networks—their inter-layer information flow and learning dynamics (Shwartz-Ziv & Tishby, 2017)—are one of its application scenarios. The fiber framework places its analytical focus on the information transmission in multi-agent socio-technical systems—from the true state, the sensor, the model, the user interface to the user's decision—and on the corresponding institutional and cognitive conditions. This is a difference in application focus, not a boundary confining the Information Bottleneck method in principle to the interior of models.

Difference Two: Value Judgment. The Information Bottleneck specifies, through the task variable and its joint distribution, which information is relevant to the target, and analyzes the trade-off between retaining relevant information and compressing the input—under appropriate conditions, discarding information irrelevant to the task may aid generalization; this formalization does not replace an institutional audit of the choice of the task itself. The dimensions judged "irrelevant" and discarded may contain tail risks, the interests of minority groups, and stability across institutional cycles. The arbiter of "irrelevant information" is the designer of the loss function, not all stakeholders of the true state.

Difference Three: Computability. The Information Bottleneck uses the mutual information $I(X;Z)$ defined on a specified probabilistic model to characterize compression (this section uniformly denotes its compressed-representation variable by Z, to avoid confusion with the T that denotes the true state in this book), and its estimation still depends on data, models, and estimation methods. The fiber framework uses the "size" of the fiber $\pi^{-1}(r)$ to measure uncertainty; in many open socio-technical scenarios it lacks sufficient information about the state space, the mapping, and the measure, and is therefore currently difficult to calculate directly—this is precisely the current core weakness of the framework and the direction the above proxy indicator schemes attempt to address. This limitation does not cancel the diagnostic value, but it constrains the precision of the current conclusions—it shares its root with the computability difference from Information Bottleneck in the differing scope of the objects analyzed: the nature of the framework determines that it is currently difficult to quantify, not that it is unwilling to be quantitative—when the object of analysis extends to socio-technical systems lacking sufficient observations and constraints, precise measurement is often more difficult, and the extent of this difficulty depends on the specific object and evidence conditions rather than on the category "socio-technical system" itself; but the necessity of diagnosis does not weaken on that account.

The relationship between the two is analogous to that between classical mechanics and thermodynamics—sharing energy conservation, but differing in the object of analysis and the problem consciousness. The Information Bottleneck provides a quantitative tool for analyzing task-relevant compression; the fiber framework connects these issues with the institutional diagnosis of pseudo-small fibers, pseudo-calibration, and cognitive debt. This connection does not presuppose confining the Information Bottleneck to the interior of models, nor does it claim that its mathematical tools cannot be used for multi-agent systems.

Lessons Absorbed from Information Bottleneck: Although the fiber framework and Information Bottleneck differ somewhat in application focus, the latter's core insight—**optimal compression is task-relevant**—can provide an important refinement dimension for the former.

The objective function of Information Bottleneck, $\min_{p(z|x)} I(X;Z) - \beta I(Z;Y)$—denoting the compressed representation by Z, to avoid confusion with the T that denotes the true state in this book—reveals a fact: which information a mapping process discards and which it retains depends on the **task variable** Y it is optimized

to predict. In the language of the fiber framework, this means: - The hidden subspace $\mathcal{T}_h$ of a system is not uniform—the information it contains includes some that is indeed irrelevant to the current task Y (under suitable conditions, discarding it may improve generalization) and some that is crucial for task Y' (a task not being optimized) (discarding it creates hidden risk). - Therefore, a more refined diagnosis should not stop at "mapping is lossy, so the hidden subspace is large," but should pursue: **"For which task Y did this mapping process optimize its information preservation? For other possible tasks Y', which potentially crucial information did it discard?"**

This line of inquiry pushes the fiber framework's diagnosis from the generalized level of "all mappings are lossy" to the refined level of "the sufficiency of a specific mapping for a specific set of tasks." One of the directions for V2.0 is: to define a set of key tasks Y' for a specific domain (such as AI alignment)—for example, factual accuracy, reasoning faithfulness, value consistency—and to develop methods for measuring the degree of information preservation of systems on these tasks. This will progressively bring the "hidden subspace" from an unobservable concept closer to a measurable object.

§8.2 Version Number and Evolution Commitment of the Framework

The fiber framework V1.0 presented in this chapter is a qualitative diagnostic tool. Its core variables—the size of the hidden subspace, the degree of pseudo-calibration, the intensity of multi-mapping recursion—have not yet been operationally measured. The evolutionary directions for V2.0 include: (1) constructing at least one observable proxy indicator for each core variable (three candidate directions for the hidden subspace have been given above); (2) testing in controlled environments the conditional associations among mapping structure, optimization objective, and pseudo-calibration (e.g., comparing the performance of different mapping schemes under comparable tasks and evaluation conditions); (3) upgrading the three transversal dimensions (political economy, physical substrate, temporal dynamics) from qualitative dimensions to quantifiable coupling relationships. The current value of the framework lies not in "having been verified" but in "being sufficiently clear to be falsifiable." The above construction schemes for proxy indicators are the first step toward pushing the qualitative framework toward quantitative testing. The second edition of this book (projected for 2027–2028) will provide preliminary operationalization tools for each of the three directions.

The three transversal dimensions—political economy, physical substrate, temporal dynamics—cross-cut the four-layer progressive structure (they neither belong to any single layer nor constitute independent layers alongside the four; nor are the dimensions themselves mutually independent—what the framework adopts here is the analytic separability of perspectives and does not claim their ontological orthogonality)[7]. This chapter focuses on the technical unfolding of the progressive structure and the emergent layer; the political economy dimension (mapping function monopoly, the capitalization of behavioral surplus, the three stages of the Discrete Leviathan) is found in Chapters 14 through 22; the physical substrate dimension (material constraints of computing power, energy consumption, planetary boundaries) is found in Chapter 32; the temporal dynamics dimension (the accumulation and phase transition of cognitive debt, the race of the three clocks) is found in Chapter 15 (Dynamic Calibration of AI with MCAF), Chapter 22 (A History of the Governance of Financial Cognitive Debt), and Chapter 31 (An MCAF Projection of Agentic AI).

whenever two properties being programmed must compose differently and yet be coordinated, we

[7]The term "cross-cut" here borrows from the concept of cross-cutting in Aspect-Oriented Programming (AOP) by Kiczales et al. (1997). That paper defines it generally as follows:

> say that they cross-cut each other (Kiczales et al., 1997)

It classifies cross-cutting properties as follows:

> properties for which the implementation cannot be cleanly encapsulated in a generalized procedure (Kiczales et al., 1997)

This framework borrows here the connotation of the analytic separability of perspectives—the four-layer progressive structure and the three transversal dimensions each have different composition rules but must be coordinated—while discarding the connotation at the level of language/tool implementation.

The clock dimension denotes, first and foremost, an operating mode that processes readings; it is not a denial of every form of deployment state. For a mainstream foundation-model inference instance without external memory, online training, or persistent state, the parameters do not accumulate irreversible experience through a single input. Conversation histories, logs, external memories, and parameter updates are deployment-level states whose persistence, deletability, and cross-period effects require separate analysis. 'No temporal depth' is used only within this qualification.

For how names themselves shape expectations of capability, see Chapter 13's Appendix A, "Naming Audit—Naming as the First Layer of Pseudo-Calibration."

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